

Curvature and Action Based Dynamics for Causal Sets

PhD Prospectus

by

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Chapter 1

INTRODUCTION

Causal set theory is a discrete theory of spacetime. It is not anything radical in that it not only hopes to reproduce the predictions of GR at large scales but it also uses conceptual insights from and within the framework of GR. It uses one such insight to build its foundation, which is that the metric only carries information about two things at each point of spacetime — the volume scale (conformal factor) and the light-cone structure. This was proven in the 1970s by Hawking, King, McCarthy and Malament, and is known as the HKMM theorem (cf. [26], p. 2). CST proposes that spacetime is a set with at most a *countably* infinite cardinality, whose elements correspond to events and have binary relations among them of being causally related or not. Its claim is that the complete spacetime geometry up to scales larger than the average spacing between the elements can be recovered from it — the number of elements that give rise to a region in the continuum corresponds to the region’s volume, and the causal relations among those elements correspond to the light-cone structure in that region.¹

In mathematical terms we say that a causal set, $\mathcal{C}$, is a partially ordered set that is locally finite, where the partial order represented by $\preceq$ dictates the causal relation of the elements, i.e., if $i \preceq j$ then j is to the causal future of i .² By local finiteness, we mean that for any $p, q \in \mathcal{C}$ the cardinality of $I(p, q) = \{i \mid p \prec i \prec q\}$ is finite; $I(p, q)$ is called the *interval* between p and q and is the discrete analog of the Alexandrov set defined for two points in

¹Just a note, this discretization is not of the type in quantum field theory where calculations are performed on a lattice in spacetime as both a calculation technique and a way to define integrals — in CST spacetime is *ontologically* discrete even at the classical level.

²In most of the literature on causal sets a sub partial order $\prec$ is used, where $i \prec j$ if $i \preceq j$ and $i \neq j$ [5].

a Lorentzian manifold as $A(x, y) := I^+(x) \cap I^-(y)$. We can define a further relation $\prec^*$, called a link, where $i \prec^* j \implies i \prec j$ and $I(i, j) = \emptyset$ which denotes that i and j have no element between them. Such a relation does not exist in the continuum because there are always an uncountably infinite number of points between any pair. A causal set can be fully described either by a links matrix, $\mathcal{L}$, by specifying whether there is a link as defined above between each pair of elements in an n element causal set as an $n \times n$ matrix or a relations matrix, $\mathcal{R}$, which can be had from the former by transitive closure³, thus they both contain the same information. An example of the former is given in Fig. 1.1.

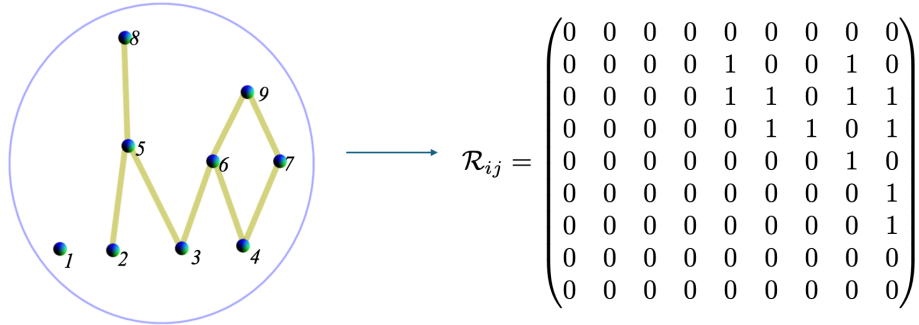


Figure 1.1: An example of a small causal set and its relations matrix. The set of elements and their relations is called a causal set. Note that the labeling of the elements is arbitrary.

Each labeling will have its own relations matrix, but all of these different matrices represent the *same* causal set, and hence the same physical setting. This reminds one of the freedom to choose arbitrary coordinates in a spacetime in general relativity. [Source: the image on the left is from [26]]

Obviously, discretizing spacetime this way raises questions about whether other key ideas of GR are upheld or violated; e.g., does the introduction of a fundamental scale, which serves to tell how many points are needed to reproduce a certain continuum volume, contradict Lorentz invariance? Over the past few decades, this and other crucial questions, specifically how some of the discrete analogs of invariant continuum quantities like the scalar curvature or even the dimension of spacetime, etc. can be formulated have been rigorously answered. These results have been described as dealing with the *kinematics* of

³Determining relations implied by transitivity, i.e., if $a \prec^* b$ and $b \prec^* c$ in the links matrix then $a \prec c$ in the relations matrix.

causal sets. The other aspect of research in CST deals with the *dynamics* of causal sets. These include both paradigms for classical as well as quantum dynamics. For CST to be a theory of quantum gravity it would have to be a quantum theory of causal sets [26] however, the quantum dynamical frameworks developed thus far are incomplete. In contrast, ideas for classical dynamics are more developed; chief among them is the framework of Classical Sequential Growth (CSG) which uses classical probabilities to grow causal sets in various ways. The main problem that remains unsolved in it is whether one can obtain from it a manifoldlike causal set.

We present two projects in this prospectus. The first one, presented in Chapter 2, deals with kinematics, and in it our goal is to find a discrete analog of the Weyl tensor for causal sets. The second project, discussed in Chapter 3, focuses on classical dynamics for causal sets, with the aim of formulating dynamics based solely on the discrete counterpart of the *action*, without relying on variation or using a Lagrangian, as is usually done in continuum-based theories. The second project is significantly less developed than the first — it is more of an exploration of why it is difficult to formulate even classical dynamics for causal sets and of ideas in continuum literature that might be useful in overcoming these problems. Having said this, the first project, of building the analog of the Weyl tensor, can benefit the second project because it might turn out that the complete dynamics of causal sets depends on both kinds of curvature, as opposed to GR in which the Einstein-Hilbert action contains only the Ricci scalar.

We would like to mention here that if this document is the reader’s first exposure to the causal set hypothesis, then it is natural for them to have questions of a general nature about CST that do not directly pertain to our work. If that is the case, I would like to point the interested reader to the review in [26]. One such question that is nonetheless relevant to our work is about the uniqueness of a causal set that can be “faithfully embedded” in a given manifold. A faithful embedding is a map $\Phi : \mathcal{C} \rightarrow M$, where $\mathcal{C}$ is the causal set and (M, g)

is the manifold it is meant to reproduce, which satisfies two properties: (i) if $x \prec y$ in the causal set, then $\Phi(x) \in J^-(\Phi(y))$, where $J^-(p)$ is the causal past of $p \in M$; (ii) $\Phi(\mathcal{C})$ is a uniform distribution in (M, g) with respect to the spacetime volume measure of (M, g) . The existence of a faithful embedding means that the causal set can reproduce a manifold in a meaningful way. It turns out that the causal set that can be faithfully embedded in a manifold is not unique; in fact, multiple causal sets can be embedded in the same spacetime (this will be easy to see when we introduce the idea of generating multiple causal sets from the same spacetime region called *sprinkling* in section 2.2.1).⁴

⁴One can ask the converse question as well: can a causal set be faithfully embedded into multiple manifolds? This is the same as asking if a causal set carries information to reproduce different spacetimes simultaneously. The claim of CST theory is that the answer is in the negative. If it were otherwise, then one would have to explain why a specific spacetime out of the many is physically realized. This is the fundamental conjecture of CST — its *Hauptvermutung*: If a single causal set faithfully embeds into two different manifolds, then the two manifolds must be isometric above the scale of average separation between its elements. This conjecture is very important for CST because it ensures that a causal set has the capacity to hold all the information about a manifold and that there is no ambiguity in the properties of the manifold. The conjecture has not been proven yet, but important progress has been made in getting closer to a proof. One of the problems in developing a proof is determining a criterion for defining the similarity or “closeness” of two Lorentzian geometries.

Chapter 2

GRAVITATIONAL WAVES IN CST

2.1 Introduction

An important part of research in *kinematics* is constructing analogs of continuum invariant quantities like the scalar curvature and the action using structures that are native to a causal set, e.g., the number of *chains* of length k — paths that connect the starting and ending elements of a subset of a causal set such that there are $k - 1$ causally related points along the way, i.e., a k -chain is a totally ordered set of the form $\{p_1, p_2, \dots, p_{k+1} \mid p_1 \prec p_2 \prec \dots \prec p_{k+1}\}$ [1]. This is understood pictorially through Figure 2.1 from Ref. [26]. An important milestone for the community was constructing the discrete analog of

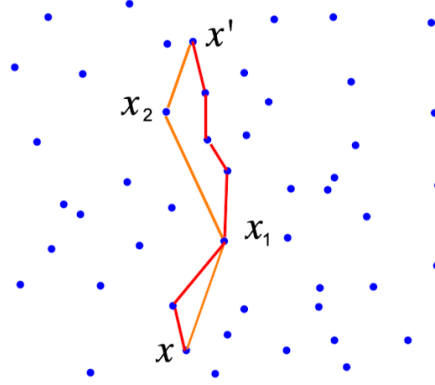


Figure 2.1: Two chains between the elements x and x' . The orange one is of size $k = 3$ and the red one is of size $k = 6$, i.e., don't count the starting element; other authors use different conventions.

the Ricci scalar, R , in terms of the number of intervals of certain cardinalities (recall that $I(p, q) = \{i \mid p \prec i \prec q\}$) because it then allowed for the construction of the Einstein-Hilbert action, which as we know is the integral of R . The full curvature of a manifold is given by the Riemann tensor, $R_{\mu\alpha\nu\beta}$, of which the Ricci tensor, $R_{\alpha\beta} = g^{\nu\mu}R_{\mu\alpha\nu\beta}$, is only

the trace part. The non-trace part is called the Weyl tensor, and its discrete analog, or that of any scalars constructed from it, however, is yet to be found. Our hope in this project is to fill this gap.

Initially, we have started with a limited class of vacuum spacetimes for which to find the Weyl tensor called plus polarized plane waves (GW_+), whose metric is the one for Minkowski spacetime with perturbations on top. These perturbations have a wave-like form characterized by the amplitude h_0 , a polarization, and a wave vector that gives direction and frequency. Since $R_{\mu\nu}$ is zero for all vacuum solutions of Einstein's equations, therefore Ricci curvature for GW_+ is identically zero, but their Weyl curvature, $C_{\mu\nu\alpha\beta}$, is not. This means that any curvature one is able to find in these spacetimes will necessarily represent the Weyl tensor. (The properties of the Weyl tensor are discussed extensively in Sec. 2.3.)

There are two stages of this project. The first is to be able to generate causal sets that represent these linearized plane waves, this is called “sprinkling” in a spacetime. We want to be able to do this so that once we have a proposal for finding the discrete analog of the Weyl curvature we will have causal sets to test it. We have successfully completed this part and its details are given in the following sections.

The second part is developing the proposal for a discrete analog of Weyl curvature itself. This part is still ongoing. We have strong reason to believe that such a proposal is achievable because there are numerous examples of successful discrete-continuum correspondences in the causal set literature starting from estimating the spacetime dimension of a causal set to the Ricci curvature, as mentioned earlier. For a somewhat exhaustive list of other discrete analogs of continuum invariant quantities, please see Section 4 of Ref. [26]. Not only do we have such proposals for different continuum invariant quantities, it turns out that there are multiple discrete analogs of the *same* continuum invariant quantities. For example, apart from the interval-based definition of the Ricci scalar, there is a “chain-based” definition as well. Since the Weyl tensor is well define and not inherently different

from these invariant quantities, there is no a priori reason to believe that its discrete analog cannot be found. Finally, if we are successful in finding such a discrete analog for the Weyl curvature our hope is to generalize our results to a more general class of spacetimes called pp-wave spacetimes.

The significance of our work is that it will provide a more holistic picture of curvature in CST. It will also illuminate what goes into dealing with dynamical spacetimes in CST. Finally, by finding an analog of the Weyl tensor we will be strengthening a restricted version of the *Hauptvermutung* which says that a causal set can embed into two different manifolds *iff* they have the same invariant quantities, $\mathcal{G} = \mathcal{G}'$, associated with a given discrete invariant $\mathcal{O}$ in the causal set. This makes sense because the same causal set should not be embeddable in two spacetimes with different Ricci scalars, for example. But for one to claim this one needs to have as many order invariants corresponding to continuum invariants as possible. Our work will add to this list by providing an analog for the Weyl tensor, thus providing a way to confirm the restricted *Hauptvermutung*.

2.2 Sprinkling in plus-polarized plane wave

2.2.1 What does sprinkling into a manifold mean?

Sprinkling points into a Lorentzian manifold means generating coordinates of points of a manifold (in a fixed frame) and imposing a “partial order” on these points based on their causal relation in the manifold, i.e., which point is in the future of which other [26]. The coordinates are generated as random numbers distributed according to the manifold’s volume measure, $\sqrt{-\det g_{\mu\nu}} \equiv \sqrt{-g}$, taken as a probability distribution function; if the measure is not a function of some of the coordinates then those coordinates are generated uniformly. This method of generating points has been called the Poisson process and it has been shown [26] that the average density of points generated this way is frame independent.

The other part of sprinkling, finding the causal relations, is not straightforward for a general manifold, however. This part of sprinkling is discussed in the next subsection.

The reason for using $\sqrt{-g}$ as the “volume measure” is that it can be shown that $\sqrt{-g}$ is equal to the Jacobian, $J = \det[\partial x^\alpha / \partial x^{\alpha'}]$, of the transformation $x^\alpha \rightarrow x^{\alpha'}(x^\alpha)$ from Cartesian ($x^{\alpha'}$) to arbitrary coordinates (x^α) [18], and since we know from differential calculus that $dV := d^4x = J d^4x' \implies dV = \sqrt{-g} d^4x'$. And to see why the distribution of points depends on the volume measure in the first place consider the example of generating spherical coordinates for points in $\mathbb{R}^3$. One immediately realizes that more points need to be generated for larger values of r because the volume element, $dV = r^2 \sin(\theta) d\phi d\theta dr$, “grows” with r unlike the situation in Cartesian coordinates, where the volume element, $dx dy dz$, is constant everywhere. For our metric, a *plus-polarized linear plane wave* (GW_+),

$$ds^2 = -dt^2 + (1 + h_{xx}) dx^2 + (1 - h_{xx}) dy^2 + dz^2 + \mathcal{O}(\frac{2}{0}),$$

where $h_{xx} = h_0 \cos[\omega(t - z)]$ and h_0 is perturbation parameter of our choosing (see discussion in a later subsection for our specific choice),

$$\sqrt{-g} = 1 - \frac{h_0^2}{2} \cos^2[\omega(t - z)] + \mathcal{O}(\frac{3}{0})$$

up to first order in h_0 . Later on we will be using null coordinates, $u := (t - z)/\sqrt{2}$ and $v := (t + z)/\sqrt{2}$, and in them $h_{xx} = h_{xx}(u)$ and

$$g_{\mu\nu} = \begin{pmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 + h_{xx}(u) & 0 \\ 0 & 0 & 0 & 1 - h_{xx}(u) \end{pmatrix}. \quad (2.1)$$

The determinant g retains its form in null coordinates except for becoming a function of u only,

$$\sqrt{-g} = 1 - \frac{h_0^2}{2} \cos^2(\sqrt{2}\omega u). \quad (2.2)$$

Thus, it acts as an unnormalized distribution for u . Formally, generating numbers according to a distribution is called sample generation or sampling. There are various ways of generating samples [11], the most well-known being the inversion method and the rejection method. We use the latter to generate the various u , one for each point, because rejection sampling works for multivariate distributions as well, which might be useful later on in the project (Fig. 2.2). On the other hand, all the v, x, y are generated uniformly because the distribution is independent of these coordinates. My code can be viewed here¹.

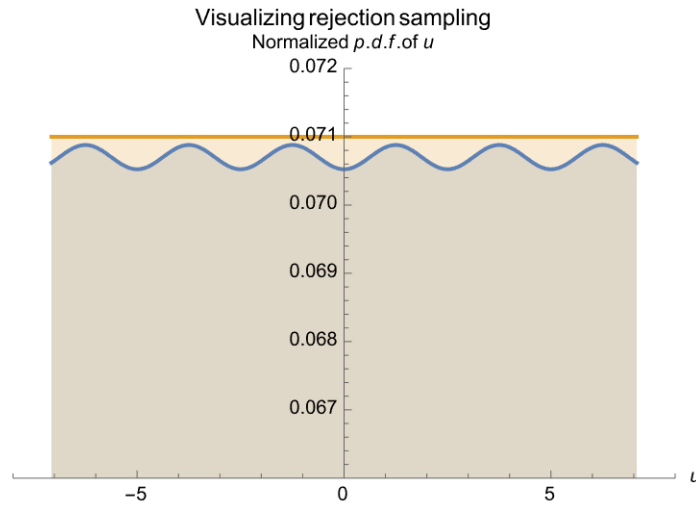


Figure 2.2: Visualizing rejection sampling and the p.d.f. for the u coordinate. The blue line is the curve of the distribution function. The orange line is the uniform reference distribution, $U_1 = c$, where $c \geq \max(\text{pdf}(u))$. A second random number, $U_2 = c \cdot \text{rand}()$, is uniformly generated for every trial u, r , which is kept if $r \leq U_2$. (In general the reference distribution is one that envelopes the target distribution more tightly than a uniform distribution and is such that we have the code to generate samples according to it, e.g., a normal or Gaussian distribution.)

2.2.2 Finding relations between points

As mentioned earlier generating coordinates for points is only half of the job of sprinkling into a spacetime. The other half is finding the causal relations between the points, i.e.,

¹https://www.icloud.com/iclouddrive/048WySnLYRI7PfGyasNgfsfRg#Linear_GW_sprinkling

which points are in the future light-cone of each point. Given the metric and coordinates of two points in a general manifold it is not easy to determine whether they are related. A relation would mean that there is a curve between the points whose tangent vector is timelike or null everywhere along it. I am grateful to Dr. Leo Stein for introducing me to the following method and the ideas therein.

One general way of finding such a curve between arbitrary points is to find the geodesic between them. A geodesic is a curve whose tangent vector is parallel transported along it (an example of a geodesic is a great circle on the surface of a sphere) and since parallel transport preserves the norm of any transported vector, the norm of the tangent vector remains constant along a geodesic. To see this mathematically, for the tangent vector u^α we would say that the geodesic equation is $u^\alpha \nabla_\alpha u^\mu = 0$ and the condition for the preservation of its norm is

$$u^\alpha \nabla_\alpha (u^\mu u^\nu g_{\mu\nu}) = 0,$$

which can be expanded into

$$g_{\mu\nu} u^\nu (u^\alpha \nabla_\alpha u^\mu) + g_{\mu\nu} u^\mu (u^\alpha \nabla_\alpha u^\nu) + u^\mu u^\nu u^\alpha \nabla_\alpha (g_{\mu\nu}) = 0.$$

Because u^α satisfies the geodesic equation the first two terms are identically zero while the third term is zero because we are using a metric-compatible connection, i.e., $\nabla_\alpha g_{\mu\nu} = 0$. Thus, the norm of the tangent of a geodesic is indeed unchanged along it. Our idea is to find the character of the tangent vector of the geodesic connecting the points, x_1^μ and x_2^μ , which we want to compare. If the norm is negative or zero then the two points are related.

To find an expression for the norm, we solve the geodesic equation between the points for the components of the tangent vector in terms of the known coordinates of the points. The norm can be found anywhere along the geodesic as its character does not change, so we chose x_1^μ for convenience. We solve the geodesic equation in terms of an arbitrary tangent

vector at x_1^μ and the coordinates of the end points and then invert the solution to make the subjects the components of the vector.

An important conceptual requirement for this method to work is that there be a unique geodesic connecting *any* two points in the region we are sprinkling into, otherwise there can be points that are related by some timelike curve that is not a geodesic of the manifold ([9], p. 5). Such a region of a manifold is called “geodesically connected” [14], p. 300. One way to ensure this is by sprinkling in a “convex neighborhood” because the *definition* of a convex neighborhood $\mathcal{C}$ is precisely that “for *any* two points p, q of an open set $\mathcal{C}$ in a semi-Riemannian manifold there is a unique geodesic segment $\sigma_{pq} : [0, 1] \rightarrow M$ from p to q that lies entirely in $\mathcal{C}$ ” ([16], p. 129). The good news is that each point of a Lorentzian manifold has a convex neighborhood ([16], p. 130). But what is the size of a “sufficiently small neighborhood” so that it can assumed to be convex? Unfortunately, this question is not addressed in most of the literature we have consulted. As a tentative answer, however, in our work we require that the region be smaller than the length scale, l_R , at which curvature manifests itself, which is usually $\sim 1/\sqrt{R_{\mu\nu\alpha\beta}}$ where $R_{\mu\nu\alpha\beta}$ is the largest component of the Riemann tensor. For the case of a plus-polarized linearized gravitational wave (GW₊) all non-zero components of the Ricci tensor have a maximum absolute value of $\frac{1}{2}\omega^2 h_0$, therefore $l_R \sim \frac{1}{\omega} \sqrt{\frac{2}{h_0}}$. Since h_0 is small, l_R is large.

Another way through which we ensure that our procedure of using geodesics to find relations between points does not exclude some pairs of points is to make sure that the region we want to sprinkle in lies in the domain of validity of the mathematical solution to the geodesic equation.

Key steps of the calculations:

We use null coordinates, $u := (t - z)/\sqrt{2}$ and $v := (t + z)/\sqrt{2}$, throughout. Let a geodesic be a curve γ given by components $w^\mu(\lambda)$, where λ is the affine parameter along γ . So,

for example, $w^u(\lambda) = u(\lambda)$, $w^x(\lambda) = x(\lambda)$ and so on, are the individual components as functions of the affine parameter. Then, the geodesic equation is actually four coupled differential equations, one for each component function:

$$\frac{d^2 w^\mu}{d\lambda^2} + \Gamma_{\alpha\beta}^\mu \frac{dw^\alpha}{d\lambda} \frac{dw^\beta}{d\lambda} = 0, \quad (2.3)$$

and solving this equation requires us to find the Christoffel symbols,

$$\Gamma_{\alpha\beta}^\mu = \frac{1}{2} g^{\mu\nu} (\partial_\alpha g_{\nu\beta} + \partial_\beta g_{\nu\alpha} - \partial_\nu g_{\alpha\beta}). \quad (2.4)$$

To find $g^{\mu\nu}$ in the expression for $\Gamma_{\alpha\beta}^\mu$ we note that it must satisfy $g_{\alpha\beta} g^{\beta\nu} = \delta_\alpha^\nu$, which it can do if the x, y components are $(1 - h_{xx})$ and $(1 + h_{xx})$ respectively. Because then, for example, the xx element of the product matrix is $1 - h_{xx}^2 \approx 1$, since higher order terms in h_0 are ignorable *throughout* the analysis. However, in fact we can use $\eta^{\mu\nu}$ instead of $g^{\mu\nu}$. For if we claim that $\partial_\alpha g_{\mu\nu}$ are of the order of h_{xx} then terms of the form $g^{\mu\lambda} \partial_\alpha g_{\mu\nu}$ have only second order terms different from $\eta^{\mu\lambda} \partial_\alpha g_{\mu\nu}$, which we can ignore.

The number of non-zero Γ s is small because most of the $\eta^{\mu\lambda} \partial_\alpha g_{\mu\nu}$ terms are zero due to the sparsity of η (obviously, represented in null coordinates) and because the $h_{\lambda\beta}$ are non zero only for h_{xx} and $h_{yy}(= -h_{xx})$. Plus, the h are only functions of u . So, the only non-zero Christoffels are:

$$\begin{aligned} \Gamma_{xx}^v &= -\Gamma_{yy}^v = \frac{1}{2} \partial_\mu h_{xx} \\ \Gamma_{xu}^x &= \Gamma_{ux}^x = \frac{1}{2} \partial_\mu h_{xx} \\ \Gamma_{yu}^y &= \Gamma_{uy}^y = -\frac{1}{2} \partial_\mu h_{xx} \end{aligned} \quad (2.5)$$

Using these we can write the four differential equations for the components of the geodesic,

$$\frac{d^2 u}{d\lambda^2} = 0 \quad \because \Gamma_{\alpha\beta}^u = 0, \quad (2.6)$$

$$\frac{d^2 v}{d\lambda^2} + \frac{1}{2}(\partial_u h_{xx}) \left[\left(\frac{dx}{d\lambda} \right)^2 - \left(\frac{dy}{d\lambda} \right)^2 \right] = 0 \quad (2.7)$$

$$\frac{d^2 x}{d\lambda^2} + (\partial_u h_{xx}) \frac{dx}{d\lambda} \frac{du}{d\lambda} = 0 \quad (2.8)$$

$$\frac{d^2 y}{d\lambda^2} - (\partial_u h_{xx}) \frac{dy}{d\lambda} \frac{du}{d\lambda} = 0 \quad (2.9)$$

We can immediately see that the equation for $u(\lambda)$ is straightforward to solve, with the solution:

$$u(\lambda) = \frac{u_2 - u_1}{\lambda_2} \lambda + u_1 \quad (2.10)$$

where, λ_2 and u_2 are the values of the affine parameter and $u(\lambda)$ at the end point x_2^μ . Similarly, u_1 is the value of $u(\lambda)$ at the base point x_1^μ where $\lambda = \lambda_1 = 0$.

Next, we note that the equations for $x(\lambda)$ and $y(\lambda)$ are the same modulo a sign so they can be solved together. Moreover, they don't contain any zeroth-order derivatives so they can be solved as first order equations for the tangent vector, $\equiv \chi_\pm$, itself — χ_+ for $dx/d\lambda$ and χ_- for $dy/d\lambda$, respectively. Finally, they are only coupled to $du/d\lambda$ which we have seen is a constant, $u_{11} := (u_2 - u_1)/\lambda_2$. Thus, the equations for x, y can be solved by solving a simple first order equation for $\chi_\pm(\lambda)$,

$$\frac{d\chi_\pm}{d\lambda} \mp u_{11} w' h_0 \sin \omega' (u_{11} \lambda + u_1) \chi_\pm = 0, \quad (2.11)$$

where $\omega' = \sqrt{2}\omega$. This has the solution, expanded up to first order in h_0 ,

$$\chi_\pm = C_\pm (1 \mp h_0 \cos \omega' (u_{11} \lambda + u_1)). \quad (2.12)$$

A few things to note about this solution. The factor $(1 \mp h_0 \cos \omega' (u_{11} \lambda + u_1))$ will always be positive for small h_0 . So, the tangent vector's x, y components, $\chi_\pm$, do not change sign on the geodesic i.e. if the geodesic is in the positive x -direction then it can never go in the

negative x -direction. Moreover, $C_{\pm}$ can be both positive and negative depending on the initial direction of the tangent vector, however this does not affect our calculation for the norm of the tangent vector at the base of each geodesic as we will see below. Moreover, for completeness' sake we note that $\chi_{\pm} = 0$ is also a solution to the equation although the probability that two sprinkled points will have exactly the same coordinates is zero.

Our ultimate goal is to find the sign of the norm of the tangent vector, W^{μ} , at the base point x_1 . To carry forward the calculation we suppose we know the components of $W^{\mu}(0) = (u_{11}, v_{11}, x_{11}, y_{11})$. (This is already true for u_{11} as we have seen above.) So, we can define $C_{\pm}$ in terms of $x_{11}(y_{11})$,

$$C_{\pm} = \frac{x_{11}(y_{11})}{(1 \mp h_0 \cos \omega' u_1)}. \quad (2.13)$$

We also want to be able to use the coordinates of the two points x_1^{μ}, x_2^{μ} so we integrate the expressions for $\chi_{\pm}$,

$$x(\lambda)(y(\lambda)) = \frac{x_{11}(y_{11})}{(1 \mp h_0 \cos \omega' u_1)} \left(\lambda \mp \frac{h_0}{\omega' u_{11}} \sin \omega' (u_{11} \lambda + u_1) \right) + c_2. \quad (2.14)$$

If we now use the values for the components of x_1^{μ}, x_2^{μ} and subtract the expressions we get

$$[\Delta x](\Delta y) \equiv [x_2 - x_1](y_2 - y_1) = \frac{x_{11} \cdot \lambda_2}{(1 \mp h_0 \cos \omega' u_1)} \left[1 \mp \frac{h_0}{\omega' \Delta u} (\sin \omega' u_2 - \sin \omega' u_1) \right], \quad (2.15)$$

$$\Rightarrow \boxed{x_{11}(y_{11}) = \frac{1}{\lambda_2} \Delta x(\Delta y) \left[1 \mp h_0 \left(\cos \omega' u_1 - \frac{1}{\omega' \Delta u} (\sin \omega' u_2 - \sin \omega' u_1) \right) \right]}. \quad (2.16)$$

Here, $\Delta u = u_2 - u_1$ and λ_2 is the affine parameter at x_2^{μ} whose value we don't know, except that it is positive. It will turn out that that in expression for v_{11} we will also have the same factor and thus the norm will have $\frac{1}{\lambda_2^2}$ multiplied to it which doesn't effect its sign.

We can use Eq. 2.11 in Eq. 2.7 to similarly solve for v_{11} . We summarize the results below. Let

$$E := \frac{1}{\omega' \Delta u} (\sin \omega' u_2 - \sin \omega' u_1) - \cos \omega' u_1 \quad \text{and,} \quad (2.17)$$

$$\Delta_{xy} := (\Delta x)^2 - (\Delta y)^2. \quad (2.18)$$

Then,

$$u_{11} = \frac{1}{\lambda_2} \Delta u, \quad (2.19)$$

$$v_{11} = \frac{1}{\lambda_2} \left[\Delta v + \frac{\Delta_{xy}}{\Delta u} \frac{h_0}{2} E \right], \quad (2.20)$$

$$x_{11} = \frac{1}{\lambda_2} \Delta x [1 + h_0 E], \quad (2.21)$$

$$x_{11} = \frac{1}{\lambda_2} \Delta y [1 - h_0 E]. \quad (2.22)$$

The norm is given by $g_{\mu\nu} W^\mu W^\nu|_{\lambda=0}$. Using the metric in Eq. 2.1 and keeping terms up to first order in h_0 we get,

$$|W(0)| = \frac{1}{\lambda_2^2} [(-2\Delta u \Delta v + (\Delta x)^2 + (\Delta y)^2) + h_0 [(\Delta x)^2 - (\Delta y)^2] \text{sinc}(\omega' \Delta u / 2) \cos(\omega' [u_2 + u_1] / 2)]. \quad (2.23)$$

(Here, we use $\text{sinc } x = (\sin x)/x$.)

Interpretation of Eq. 2.23

We can use Eq. 2.23 to find the character of the geodesic between any two points that we generate the coordinates for: if it is negative then the second point is in the future of the first. (The expression itself does not reveal the ordering in time because if we exchange point 1 with 2 the expression remains unchanged, however, in our code we first find out

the time ordering of the points using that fact $t = (u - v)/2$.) The expression on the first line is the Minkowski norm between the points; it is only the expression on the second line that reveals the effect due to the gravitational wave perturbation. This second term itself is comprised of three terms: the magnitude of the first one is maximum for either $\Delta x = 0$ or $\Delta y = 0$ and zero for $\Delta x = \Delta y$ — this is probably due to the wave being plus-polarized which means that there is little deformation in the cross ($\times$) directions; the $\text{sinc}(\omega' \Delta u/2)$ term is maximum, one, at $\Delta u = 0$, i.e., there is maximum deviation from the Minkowski interval if the two points have the same phase $u = (t - z)$; the cosine term is a modulation term that mostly dictates the sign of the deviation from the Minkowski interval.

2.2.3 Default values for the variables

The upper limit of the number of points that can be numerically sprinkled on a personal computer in realistic times, ($\sim 1\text{min}$), is $N = 10,000$. In 4D this corresponds to 10 points in each direction at each “time-slice”, thus when choosing the volume to sprinkle into the size of each coordinate domain $:= \Delta\kappa$, ‘timeinterval’ in my code, should not be more than ~ 10 in whatever units we want to simulate. And if we want at least $n_W = 2$ full waves to be captured in each of the slices, i.e., $\lambda = \frac{\Delta t}{n_W}$, then the frequency of the plane wave $\omega = \frac{2\pi}{\lambda} = \frac{2\pi n_W}{\Delta t} = \frac{2\pi}{5}$ ($c = 1$).

To find an appropriate value for h_0 we recall an earlier discussion where we explained that we would like the size of the region of sprinkling to be smaller than the curvature length scale, $l_R \sim \frac{1}{\sqrt{R_{\mu\nu\alpha\beta}}}$, where $R_{\mu\nu\alpha\beta}$ is the largest component of the Riemann tensor. For the case of GW_+ all non-zero components of the Ricci tensor have an absolute value of $\frac{1}{2}\omega^2 h_0 \cos \omega(t - z)$ therefore, $l_R \sim \frac{1}{\omega'} \sqrt{\frac{2}{h_0}}$. Thus, if we want $\Delta\kappa$, the size of our region,

$< l_R$ then, since $\omega' = \frac{2\pi n_W}{\Delta t}$,

$$\begin{aligned} \frac{\Delta t}{2\pi n_W} \sqrt{\frac{2}{h_0}} &> \Delta t. \\ \implies h_0 &< \frac{1}{2\pi^2 n_W^2}. \end{aligned} \tag{2.24}$$

That is, the maximum tolerable value for h_0 to remain within l_R , $h_0^{\max}$, only depends on the number of waves we want to sprinkle into. This makes intuitive sense because sprinkling into less than a wavelength, $n_W < 1$, the curvature would vary very little over the domain. For $n_W = 2$, $h_0^{\max} \sim 0.01$. To see appreciable results, i.e., deviations from Minkowski, we use this value for h_0 . This is in contrast to what we find physically for typical gravitational waves where $h_0 < \sim 10^{-22}$. This departure from observation is similar to interpreting the *scale of separation* between the sprinkled elements because, technically, the separation between the elements in a causal set is expected to be of the order of the Planck length, $l = \sqrt{\hbar G/c^3} \sim 10^{-35}\text{m}$. This would mean, using the numbers in the first paragraph of the subsection, that we are saying that n_W number of waves can fit in a 10 Planck length region.

However, these departures are due to practical limitations of being able to sprinkle large values of N . For example, to sprinkle one wavelength of a km scale wave with Planck length separation between points, $N \sim (10^{38})^4 = 10^{152}$, and the number of comparisons needed to determine the causal relations between all the elements, which scales as N^2 , would be $\sim 10^{304}$. So, the causal sets we are generating should be interpreted as having large element separation, that is, they don't represent the smallest scale structure of spacetimes. Such causal sets are called “coarse-grained”. Such causal sets are expected to nonetheless contain discrete analogs of continuum features larger than the scale of separation between causal set elements, thus, since the Weyl curvature changes over scales larger than l_R we should be able to recover its discrete counter-part at large scales.

2.3 Weyl Tensor

The Riemann tensor for a manifold $(\mathcal{M}, g)$ contains all the information about the curvature at each point. It can, however, be separated into trace and trace-free parts. The trace part is given by the Ricci tensor $R_{\mu\nu} := R^\rho{}_{\mu\rho\nu}$ (all other contractions of the Riemann tensor are either 0 or, because of its symmetries, related to the Ricci tensor). The non-trace part is given by the Weyl tensor. This is the part of the curvature that is not specified by the sources directly (there is nonetheless a differential relation which is discussed below) because Einstein's equation, $R_{\mu\nu} = 8\pi(T_{\mu\nu} - \frac{1}{2}Tg_{\mu\nu})$, only relates the Ricci tensor to the energy-momentum tensor. The analog of this kind of source independence is seen in electromagnetism where the field in a region is both that given by the charges and currents everywhere in space *and* that given as part of the setting, for example, an electromagnetic wave passing by which need not have a source [2].

In n dimensions the Weyl tensor is [7]

$$C_{\rho\sigma\mu\nu} = R_{\rho\sigma\mu\nu} - \frac{2}{n-2}(g_{\rho[\mu}R_{\nu]\sigma} - g_{\sigma[\mu}R_{\nu]\rho}) + \frac{2}{(n-1)(n-2)}g_{\rho[\mu}g_{\nu]\sigma}R. \quad (2.25)$$

The $[\]$ in the subscript mean anti-symmetrization. The expression in Eq. 2.25 ensures that all the contractions of $C_{\rho\sigma\mu\nu}$ are zero while it retains the symmetries of the Riemann tensor [7],

$$C_{\rho\sigma\mu\nu} = C_{[\rho\sigma][\mu\nu]}, \quad C_{\rho\sigma\mu\nu} = C_{\mu\nu\rho\sigma}, \quad C_{\rho[\sigma\mu\nu]} = 0. \quad (2.26)$$

Some of its significant properties and their relevance to our project are given below:

- i. The Weyl tensor is defined in three or more spacetime dimensions only, and it can be shown that it is zero in three dimensions [7]. This implies that gravitational waves, which have non-zero Weyl curvature, must occur in 4D. Hence, our code is also in 4D.

- ii. In 4D, there are ten independent components of the Weyl tensor. The Riemann tensor overall comprises twenty components, with the Ricci tensor accounting for the other ten.
- iii. In vacuum, the Weyl tensor is equal to the Riemann tensor because Einstein's equation implies that $R_{\mu\nu} = 0$ if $T_{\mu\nu} = 0$, so, only the Weyl part of the Riemann tensor is nonzero. This is important for our work in building a discrete analog of the Weyl tensor because it allows us to apply the geodesic deviation equation to causal sets as a means to retrieve the components of the Weyl tensor, even though the equation is expressed in terms of the Riemann tensor's components. This is outlined as a proposal later on.

For reference, we give below the Weyl (= Riemann) tensor for plus-polarized plane GWs (GW_+) and the field outside a spherical mass in the Newtonian limit (NL) of Einstein's equations. Both are weak gravitational fields in vacuum and appear as perturbations on top of the Minkowski metric.

$$\begin{aligned}
C_{x0x0}^+ &= -C_{y0y0}^+ = -C_{x0xz}^+ = C_{y0yz}^+ = C_{xzzx}^+ = C_{yzyz}^+ \\
&= \frac{1}{2}\omega^2 h_0 \cos \omega(t - z),
\end{aligned} \tag{2.27}$$

and

$$\begin{aligned}
C_{\alpha ij\alpha}^{\text{NL}} &= -\frac{\partial}{\partial x_i} \frac{\partial}{\partial x_j} \left(\frac{\phi_0}{r} \right) \quad \text{for } \alpha = t, x, y, z, \\
&= (-3x_i x_j + \delta_{ij} r^2)/r^5
\end{aligned} \tag{2.28}$$

Here, ϕ_0 is the constant potential at the surface of a spherical mass at the center of the coordinate system. For both of them, the rest of the $C_{\alpha\beta\mu\nu}$ can be found by applying the symmetries given in Eq. 2.26 on Eq. 2.27 and Eq. 2.28 respectively.

- iv. The Riemann tensor obeys the so-called second or differential (as opposed to algebraic) Bianchi identity:

$$\nabla_{[\lambda} R_{\rho\sigma]\mu\nu} = 0.$$

This identity implies a first-order differential relation for the Weyl tensor in terms of the covariant derivatives of the Ricci tensor and scalar, which in turn can be written in terms of the energy-momentum tensor, $T_{\mu\nu}$, through the Einstein equation, giving us (p. 169 of [7]),

$$\nabla^\rho C_{\rho\sigma\mu\nu} = 8\pi G \left(\nabla_{[\mu} T_{\nu]\sigma} + \frac{1}{3} g_{\sigma[\mu} \nabla_{\nu]} T \right).$$

Thus, for a given energy-momentum distribution, there are many solutions for the components of the Weyl tensor, each one determined by a boundary condition. For example, both plane GWs and the weak-field Newtonian approximation are vacuum solutions to the Einstein equation, yet they have different Weyl tensors, and we see that the behaviour of their components at spatial infinity is different: idealized plane waves keep oscillating, whereas the latter go to zero as $1/r^3$, thus approaching flat Minkowski spacetime.

- v. One of its most important properties is invariance under conformal transformations. This means that if one calculates $C^\rho{}_{\sigma\mu\nu}$ using some metric $g_{\mu\nu}$ and again using a transformed metric $\omega^2(x)g_{\mu\nu}$, one gets the same tensor. Here $\omega(x)$ is some non-vanishing function of spacetime coordinates [7]. In the context of causal sets, this property makes work on the Weyl tensor difficult because finding relations between elements in a non-conformally flat metric is not straightforward.

2.3.1 Geometrical interpretation

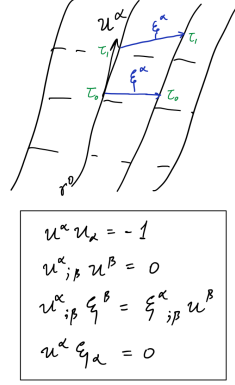
There is a difference between the effects on timelike or null geodesics of regions that have either purely Ricci or Weyl curvature. In a Blog post Dr. John Baez of the UC River-

side explains the difference through an example of a sphere of freely falling particle in the presence of the two kind kinds of curvature [2]. Imagine a small round ball of comoving coffee grounds centered at a point P with the coffee ground that actually goes through P having velocity v at that instant. “The Ricci tensor R_{ab} only keeps track of the change of volume of this ball. Namely, the second time derivative of the volume of the ball is $-R_{ab}v^av^b$ times the ball’s original volume. The Weyl tensor tells the REST [sic] of the story about what happens to the ball. More precisely, it describes how the ball changes shape, into an ellipsoid.”

This interpretation can be made more precise using the idea of the evolution of geodesic congruences. Specifically, in vacuum spacetimes the Weyl tensor alone determines the evolution of something called the shear tensor, $\sigma_{\alpha\beta}$, which in turn determines the spatial shape of a group of freely-falling particles. The same idea can be analyzed in the Newman-Penrose formalism where the evolution equations are called the Sachs equations [12]). Indeed, Professor Roger Penrose has compared the influence of Weyl-curved spacetimes on null geodesics to the way an astigmatic lens affects null rays in optics [12]. We briefly present the former for timelike geodesics because those can be modeled more easily in causal sets.

Consider the setup in Figure 2.3:

We would like to know the second rate-of-change of ξ^α w.r.t. τ on the reference geodesic γ^0 . The first derivative $(\xi^\alpha_{;\beta}u^\beta)$ need not be a consequence of curvature alone as, for example, radial geodesics of Minkowski (straight lines) can fly apart in which case $\xi^\alpha_{;\beta}u^\beta$ would be non-zero, but it turns out that the second derivative $(\xi^\alpha_{;\beta}u^\beta)_{;\sigma}u^\sigma$ would be zero due to the absence of curvature as we will see below. Independence from the first derivative might, in fact, be crucial to constructing the causal set analog of the Weyl curvature because it is difficult to select geodesics that have the same direction in causal sets since geodesics themselves are defined uncharacteristically as longest *maximal chains*. The first derivative



Let $\mathcal{O}$ be an open region in spacetime.
A congruence in $\mathcal{O}$ is a family of curves
s.t. through each point in $\mathcal{O}$ there passes
one and only one curve from this family.

u^α : tangent vector field to the cong.

ξ^α : deviation vector field
(tangent to the cross-curves)

τ : parameter along the geodesics

Figure 2.3: The solid black lines denote timelike geodesics that all have the same tangent vector u^α . The dashed curves cross the geodesics at the same value of proper time and represent a deviation vector field given by ξ^α , which is orthogonal to u^α [18].

however allows us to introduce the vector field

$$B_{\alpha\beta} = u_{\alpha;\beta}$$

which, from $\xi^\alpha_{;\beta} u^\beta = u^\alpha_{;\beta} \xi^\beta$, since $B_{\alpha\beta}$ determines the first derivative:

$$\xi^\alpha_{;\beta} u^\beta = B^\alpha_{\beta} \xi^\beta.$$

This tensor is orthogonal to u^α , as $u^\alpha B_{\alpha\beta} = u^\alpha u_{\alpha;\beta} = \frac{1}{2}(u^\alpha u_\beta)_{;\beta} = 0$ and $B_{\alpha\beta} = u_{\alpha;\beta} u^\beta = 0$ [18]. It can be decomposed into trace, symmetric-tracefree, and antisymmetric parts, all of which will also be transverse:

$$B_{\alpha\beta} = \frac{1}{3}\theta h_{\alpha\beta} + \sigma_{\alpha\beta} + \omega_{\alpha\beta},$$

where θ is the trace of $B_{\alpha\beta}$ and is called the expansion scalar, $\sigma_{\alpha\beta} = B_{(\alpha\beta)} - \frac{1}{3}\theta h_{\alpha\beta}$ is the shear tensor that we are interested in, and $\omega_{\alpha\beta} = B_{[\alpha\beta]}$ is the rotation tensor.

The interpretation of these quantities comes from the Newtonian theory of deformable media [18]. Let $\xi^a(t)$ be a small displacement between two points in a 2 D membrane then, taking the first point as a reference up to the first order of the displacement we must have

$$\frac{d\xi_a}{dt} = B^a_b(t)\xi^b + \mathcal{O}(\xi^2),$$

for some 2×2 matrix B^a_b , which for now only shares the name of $B_{\alpha\beta}$ defined above. Then for short time intervals the separation between the points on the membrane, ξ^a , would change by

$$\Delta\xi^a = \xi^a(t_1) - \xi^a(t_0) = B^a_b(t_0)\xi^b(t_0)\Delta t + \mathcal{O}(t^2),$$

where $\Delta t = t_1 - t_0$ [18]. If we suppose that $\xi^a(t_0) = r_0(\cos \phi, \sin \phi)$, i.e., a circle on the membrane of radius r_0 and B^a_b is symmetric and tracefree, i.e., a pure shear tensor,

$$B^a_b = \begin{pmatrix} \sigma_+ & \sigma_\times \\ \sigma_\times & -\sigma_+ \end{pmatrix}, \quad (2.29)$$

(here $\sigma_+, \sigma_\times$ are numbers, not matrices) then

$$\Delta\xi^a = r_0\Delta t((\sigma_+ \cos \phi + \sigma_\times \sin \phi), (-\sigma_+ \sin \phi + \sigma_\times \cos \phi))$$

and [18],

$$r_1(\phi) = \|\xi^a(t_1)\| = r_0(1 + \sigma_+\Delta t \cos 2\phi + \sigma_\times\Delta t \sin 2\phi).$$

We can relate this change quite well to our GW_+ by taking $\sigma_\times = 0$ and plot the deformed ring in the 2D membrane at t_1 , which is the curve given by $r_1(t_1)$:

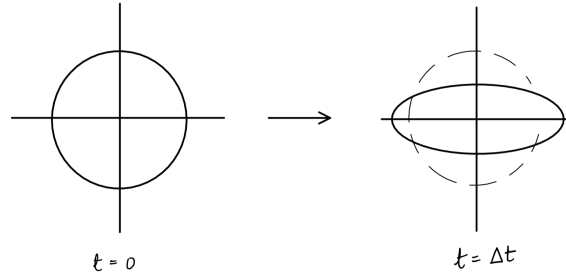


Figure 2.4: Effect of the plus shear tensor on a ring in a 2D membrane for a short interval Δt . We see that it becomes an ellipsoid

For this case the semi-major and minor axis of the ellipse are given by $a = r_0(1 + \sigma_+\Delta t)$ and $b = r_0(1 - \sigma_+\Delta t)$, thus the area of the ellipse $A = \pi a b = \pi r_0^2 + \mathcal{O}(\Delta t^2)$ remains constant to the first order. In GR the mathematics is similar to the Newtonian case, so the

interpretation of B_b^a carries over to $B_{\alpha\beta}$ whose symmetric, trace-free part is again defined as the shear, which as above, determines the spatial shape of a ball of test particles. And as above, the volume of a freely-falling ball is unaffected by the action of the shear tensor.

To know what happens at longer times we must know the dynamics of the shear tensor. In the context of GR this means evaluating $\sigma_{\alpha\beta;\mu}u^\mu$. To do this the key result is that the derivative of $B_{\alpha\beta}$ along the congruence is equal to terms that are known, namely, $B_{\alpha\beta}$ itself and the Riemann tensor [18]:

$$B_{\alpha\beta;\mu}u^\mu = -B_{\alpha\mu}B^\mu{}_\beta - R_{\alpha\mu\beta\nu}u^\mu u^\nu.$$

Using this result it can be shown that [18]

$$\begin{aligned} \sigma_{\alpha\beta;\mu}u^\mu = & -\frac{2}{3}\theta\sigma_{\alpha\beta} - \sigma_{\alpha\mu}\sigma^\mu{}_\beta - \omega_{\alpha\mu}\omega^\mu{}_\beta + \frac{1}{3}(\sigma_{\alpha\mu}\sigma^{\alpha\mu} - \omega_{\alpha\mu}\omega^{\alpha\mu})h_{\alpha\beta} \\ & - C_{\alpha\mu\beta\nu}u^\mu u^\nu + \frac{1}{2}R_{\alpha\beta}^{\text{TT}}, \end{aligned} \quad (2.30)$$

where $h_{\alpha\beta} = g_{\alpha\beta} + u_\alpha u_\beta$ is the transverse metric which is purely spatial: $u^\alpha h_{\alpha\beta} = 0$, and $R_{\alpha\beta}^{\text{TT}} \equiv R_{\alpha\beta}^{\text{T}} - \frac{1}{3}(h^{\mu\nu}R_{\mu\nu}^{\text{T}})h_{\alpha\beta}$ and $R_{\alpha\beta}^{\text{T}} \equiv h_\alpha{}^\mu h_\beta{}^\nu R_{\mu\nu}$. The important thing for us, that wraps up this discussion, is that Eq. 2.30 makes precise the geometrical interpretation of the Weyl tensor, specifically for vacuum spacetimes for which $R_{\mu\nu}$ is zero: it determines the evolution of the shear tensor along the geodesic congruence (the other terms on the R.H.S. of Eq. 2.30 can all be found in terms of $u_{\alpha;\beta}$ for a given metric, and a known u_α , the tangent to the congruence).

2.3.2 Example of constructing R using a chain distribution in a RNN in 2D

In this subsection we will present results from Ref. [17] where the discrete analog of the Ricci scalar, R , is found using the distribution of k -chains for causal sets that correspond to Riemann normal neighborhoods (RNN) of any curved 2D spacetime as a way of show-

ing the type of calculations that are done in CST literature. The difference between this calculation and the kind we would be doing is that it is in 2D and since all 2D spacetimes are conformally flat, integration over regions in 2D spacetimes can be greatly simplified by transforming to coordinates that make the region “straight”. Before specializing to 2D and a RNN, we would like to present the concept behind this calculation applicable to higher dimensions starting with the idea of chains and Alexandrov sets.

Chains, as has already been defined in the Introduction, are collections of elements, each related to the other in temporal succession. Mathematically, a k -chain is a totally ordered set of the form $\{p_1, p_2, \dots, p_{k+1} \mid p_1 \prec p_2 \prec \dots \prec p_{k+1}\}$ [1]. An Alexandrov set, on the other hand, is the intersection of the timelike (causal) future and past of two points (say x and x' from Figure 2.1) in the continuum spacetime. Formally, $A[x, x'] := I^+(x) \cap I^-(x')$.

Now, we imagine that we sprinkle many causal sets into the same Alexandrov set in the continuum. Each sprinkling will be different because, as we have seen, the process of picking points is random. Thus, each sprinkling will have some distribution of chain lengths that will be slightly different from the distribution of another sprinkling, and so we can find the expected number of k -chains, $\langle c_k \rangle$, for this ensemble.

To relate this discussion with the curvature we ask ourselves, if we have another Alexandrov set of the same volume but in a different spacetime and we sprinkle a new ensemble of causal sets into it, how will $\langle c_k \rangle$ for this new ensemble differ from the old one? We can intuit that the difference will depend on the difference between the curvatures of the Alexandrov sets. We can see hints of this in the results of the previous subsection: if we take Minkowski spacetime as the reference spacetime and GW_+ as the other spacetime then we saw that to find the relations between points in GW_+ we had to solve the geodesic equation which depended on the Christoffel symbols, which we know determine the curvature through the Riemann tensor. And how the points are related to each other in turn will affect

how many chains we find in a causal set. Using this reasoning, or otherwise, we would also expect the shape of the Alexandrov set to depend on the curvature. Having established that we expect the curvature to affect $\langle c_k \rangle$ we might want to know the individual effects of the Ricci and Weyl curvature on $\langle c_k \rangle$. This is indeed an important question but it requires detailed attention. Our hope is that finding a discrete analog of the Weyl curvature will help us answer it.

We can now present the outline of a calculation done to find the Ricci scalar for 2D spacetimes in a Riemann normal neighbourhood. The idea is that $\langle c_k \rangle$ for an ensemble of causal sets sprinkled into the *same* Alexandrov set in the continuum equals the probability of picking $k - 1$ points successively from the Alexandrov set such that each new point is in the future of the previous one. If one accepts this fact then the rest of the steps are calculations.

Outline of calculation

If one were to sprinkle n points in an Alexandrov set, A_0 , of an arbitrary d -dimensional spacetime then the probability of picking the first point of the k -chain in a differential volume element $d^d x_0$ around some coordinate point x_1 in V_0 is $\rho_0 \sqrt{-g(x_1)} d^d x_1$, where V_0 is the volume of A_0 , with minimal element x_0 and maximal element x_k and $\rho_0 = n/V_0$ is the overall density of sprinkled points. Following that, the probability of finding the second element in a differential volume element $d^d x_1$ around a coordinate point x_2 in the Alexandrov set between x_1 and x_k is $\rho_1 \sqrt{-g(x_2)} d^d x_1$, where $\rho_1 = (n - 1)/V_0$ (because one point out of n has already been sprinkled). And so on for the rest of the $k - 3$ points with $\rho_{k-2} = [n - (k - 1)]/V_0$. That is, the probability of picking $k - 1$ points in volumes $\sqrt{-g(x_1)} d^d x_1, \sqrt{-g(x_2)} d^d x_2, \dots, \sqrt{-g(x_{k-1})} d^d x_{k-1}$ around some points $x_1, x_2, \dots, x_{k-1}$ respectively during the sprinkling process such that each x_{i+1} is in the

Alexandrov set $A_i = I^+(x_i) \cap A_0$ is given by

$$\rho_0 \sqrt{-g(x_1)} d^d x_1 \rho_1 \sqrt{-g(x_2)} d^d x_2 \cdots \rho_{k-2} \sqrt{-g(x_{k-1})} d^d x_{k-1}. \quad (2.31)$$

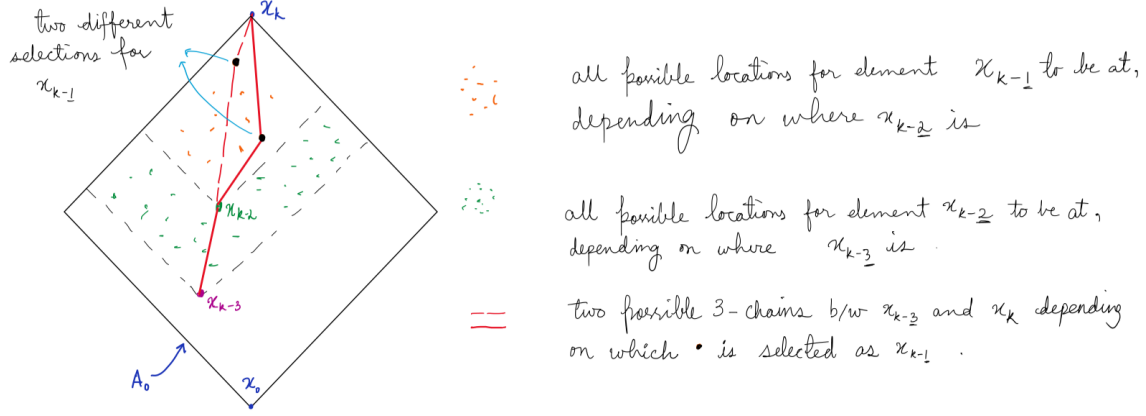


Figure 2.5: Illustration of what the different regions are and how the x_i are related to them.

To use this result in the calculation of the mean number of k -chains $\langle c_k \rangle$ in A_0 , we rely on the observation that through the volume elements $\sqrt{-g(x_1)} d^d x_1, \sqrt{-g(x_2)} d^d x_2, \dots, \sqrt{-g(x_{k-1})} d^d x_{k-1}$ there can be either no chain or 1 chain (the probability that there be 2 or more chains is a higher-order infinitesimal), so the mean number of chains through those volume elements is $0 \cdot \text{Prob}(\text{no chain}) + 1 \cdot \text{Prob}(\text{1 chain})$, and therefore equals the expression in 2.31. If we then take integrals over the coordinates of each point in the respective Alexandrov sets in (2.31) we get $\langle c_k \rangle$,

$$\begin{aligned} \langle c_k \rangle = \int_{A_0} \rho_0 \sqrt{-g(x_1)} d^d x_1 \int_{I^+(x_1) \cap A_0} \rho_1 \sqrt{-g(x_2)} d^d x_2 \cdots \\ \cdots \int_{I^+(x_{k-2}) \cap A_0} \rho_{k-2} \sqrt{-g(x_{k-1})} d^d x_{k-1}. \end{aligned} \quad (2.32)$$

(The interested reader can find a mathematically rigorous reasoning behind this identification in Meyer (1988) [15], p. 49.)

In order to solve these integrals we come back to a point alluded to previously that the shape of the A_i s depends on the curvature. One would like to “straighten” the light-cone boundaries as in Minkowski space. This is done by using the property of 2D spacetimes mentioned previously that they are all conformally flat. We can now list the key equations of the calculation.

In Riemann normal coordinates up to first order in curvature [6],

$$g_{\mu\nu} = \eta_{\mu\nu} - \frac{1}{3}R_{\mu\alpha\nu\beta}(x_0)(x - x_0)^\alpha(x - x_0)^\beta. \quad (2.33)$$

In 2D up to first order in curvature at a point in RNC,

$$R = g^{\mu\nu}g^{\alpha\beta}R_{\alpha\mu\beta\nu} = \eta^{\mu\nu}\eta^{\alpha\beta}R_{\alpha\mu\beta\nu} = -2R_{txtx}.$$

This means that the metric (2.33) will have terms in R and so will $\sqrt{-g}$. This is the first place where the curvature enters in the calculation of the integral (2.32). The second place is in finding the conformal transformation that will make the metric conformally flat in order to find coordinates in which the shape of the A_i s is “straight”. Such a transformation is found to be:

$$x = \frac{u - v}{\sqrt{2}} + \frac{R}{12\sqrt{2}}((v - v_0)(u - u_0)^2 - (v - v_0)^2(u - u_0)), \quad (2.34)$$

$$t = \frac{u + v}{\sqrt{2}} + \frac{R}{12\sqrt{2}}((v - v_0)(u - u_0)^2 + (v - v_0)^2(u - u_0)). \quad (2.35)$$

In these coordinates the line element becomes

$$ds^2 = [-2 - R(v - v_0)(u - u_0)]dudv, \text{ which is of the form} \quad (2.36)$$

$$ds^2 = \omega^2(u, v)dudv. \quad (2.37)$$

(Recall that $dudv$ is the Minkowski line element in null coordinates.) Consequently from this line element the determinant is

$$\sqrt{-g(u, v)} = 1 + \frac{R}{2}(v - v_k)(u - u_k), \quad (2.38)$$

where the expansion is defined w.r.t. the point x_k , the maximal point of the Alexandrov set, with coordinates (v_k, u_k) . Then,

$$\langle c_k \rangle = \frac{n!}{V^{k-1}(n - (k - 1))!} \int_{A_0} du_1 dv_1 \left(1 + \frac{R}{2}(v_1 - v_k)(u_1 - u_k) \right) \quad (2.39)$$

$$\int_{A_1} du_2 dv_2 \left(1 + \frac{R}{2}(v_2 - v_k)(u_2 - u_k) \right) \quad (2.40)$$

$$\dots \int_{A_{k-2}} du_{k-1} dv_{k-1} \left(1 + \frac{R}{2}(v_{k-1} - v_k)(u_{k-1} - u_k) \right), \quad (2.41)$$

where V is the volume of A_0 and n is the number of points in each sprinkling. Keeping terms only up to first order in R this equals [17]

$$\langle c_k \rangle = \frac{n!}{(n - (k - 1))!} \left[\frac{1}{((k - 1)!)^2} + \frac{NR}{\rho} \left(\frac{2k^3 - 3k^2 + k}{12(k!)^2} - \frac{k - 1}{8((k - 1)!)^2} \right) \right]. \quad (2.42)$$

This can be inverted to find R for any given chain length, thus for a single causal set there will be multiple estimates of R , one for each counting of k -chains. For $k = 3$, e.g.,

$$R = \frac{36\rho}{n} \left(\frac{4\langle c_3 \rangle}{n(n - 1)} - 1 \right). \quad (2.43)$$

Even though the expression is exact only if one uses the mean value for c_3 for an ensemble of causal sets all faithfully embeddable in the same spacetime one can nonetheless use it for the value of c_3 for a single causal set, thus getting an estimate for the scalar curvature of that causal set. Moreover, it may seem odd that a single causal set has multiple estimates of curvature, however, in the continuum limit, $\rho \rightarrow \infty$, all the estimates for R converge to the same continuum value of R .

Our goal is to find a similar calculation for the Weyl tensor, or any of its components, but it will have to be in 4D because, as remarked earlier, the Weyl tensor is nonzero only for four dimensions or greater. A brief proposal for such a construction is presented below, the details of which have not been worked out yet; this is the main part of our future work.

2.4 A sketch proposal for finding the Weyl tensor using spatial distance estimators in causal sets

Having seen an example of how to construct a discrete analog of the Ricci scalar, we present here a proposal for doing the same for the Weyl curvature. For context, we would like to say that constructing a discrete Weyl tensor is considerably harder. Volumes and timelike lengths are easy to identify in a CS as the number of elements corresponding to a region and the lengths of maximal longest maximal chains, respectively. Consequently, an expression was developed early in the history of CST that relates the Ricci scalar, volume, and timelike distances in a Riemann normal neighborhood. However, as we have seen above, the Weyl curvature has to do with spatial shape on hypersurface, and spatial information in causal sets, including shapes and length, is hard to deal with because causal sets contain information about spacetime exclusively through the relation between elements, and those are defined *in time*, i.e., spatial features at an instant of time without reference to the past or future are almost impossible to construct. This issue also makes it hard to define a Lagrangian and a Hamiltonian for a causal set, and we discuss it in the next chapter on Dynamics. Despite this, there are ways of estimating spatial distances, and in this section we propose using one such proposal in the context of geodesic deviation in causal sets to estimate a component of the Weyl tensor.

We have discussed how the evolution of the shear tensor for geodesic congruences is related to the Weyl tensor. In the context of causal sets it is harder to define congruences than individual geodesics, so a simpler approach to finding the causal set analog of the Weyl curvature might be to work with the geodesic deviation equation which describes the evolution of the separation between two neighbouring geodesics in terms of the Riemann tensor. And since the Riemann tensor for vacuum spacetimes is equal to the Weyl tensor, the geodesic deviation in GW_+ would depend on the Weyl tensor. The deviation equation

relates the acceleration of the deviation vector ξ_α between two neighbouring geodesics with the curvature as so:

$$\frac{D^2 \xi^\alpha}{d\tau^2} := (\xi^\alpha{}_{;\mu} u^\mu)_{;\nu} u^\nu = -R^\alpha{}_{\lambda\mu\nu} u^\lambda \xi^\mu u^\nu. \quad (2.44)$$

In CST, geodesics between two elements correspond to the longest maximal chain. A *maximal chain* (path) is a k -chain of the form $\{p_1, p_2, \dots, p_{k+1} \mid p_1 \preceq^* p_2 \preceq^* \dots \preceq^* p_{k+1}\}$ [1]. Furthermore, a spatial hypersurface can be defined most naturally as an inextendible antichain in causal sets because it is defined as a subset $\mathcal{A} \subset \mathcal{C}$ such that all elements $a \in \mathcal{A}$ are not related to each other and all elements of $\mathcal{C}/\mathcal{A}$ are related to some element in $\mathcal{A}$. This separates the causal set into the past and future of $\mathcal{A}$, and $\mathcal{A}$ itself just like a hypersurface does to a manifold.² So, to track the acceleration of geodesics in a causal set we will have to measure the separation between two longest maximal chains that start and end on the same inextendible antichains, corresponding to two geodesics extended between two spatial hypersurfaces Σ_{t_1, t_2} , at various points along their length. These values for separation can then be used to find a numerical estimate of their second derivative w.r.t. the proper time along the geodesics, giving us an estimate for one of the components of the Weyl tensor.

The spatial distance is hard to measure in CST. There are a few proposals in the literature to estimate it each with its own caveats. A recent one by Boguñá and Krioukov [4] although defined for distances in causal sets that are only faithfully embeddable in Minkowski spacetime nonetheless seems promising to us because of its straightforward application and their claim that it can be generalized to spacetimes that are close to Minkowski as ours is. They use the idea of *causal overlap* $\mathcal{O}(a, b) \in [0, 1]$ between points (a, b) to de-

²However, there are reasons to question this identification one of which is that these antichains do not contain all the information about their pasts like spatial hypersurfaces do, so some authors have proposed using a “thickened” antichain instead.

fine the spatial distance between them. In a causal set $\mathcal{C}$ it is defined as

$$\mathcal{O}_{\mathcal{C}}(a, b) \equiv \frac{N[C]}{N[A] + N[C]}, \quad (2.45)$$

where $N[X]$ is the number of points in the region X , and regions A and C are defined with respect to a point c in the common past of (a, b) such that they are both approximately at an equal proper time τ_c from it (Fig. 2.6):

$$A = I(a, c) \setminus I(b, c), \quad C = I(a, c) \cap I(b, c),$$

where $I(x, y) = \{e \mid x \prec e \prec y\}$. Then, for the case where $\tau_c \gg d_{\mathbb{H}^d}(a, b)$, the hyperbolic

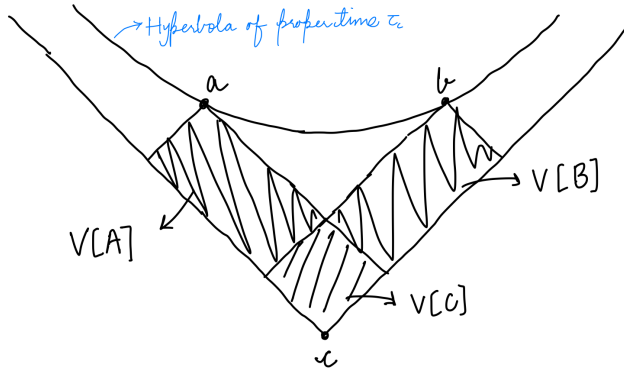


Figure 2.6: Sketch of causal overlap in 2D Minkowski spacetime. The lines emanating from each point are null rays. Region $A(B)$ is the part of the Alexandrov set between c and $a(b)$ with the past of $b(a)$ removed. Region C is common part of the future of c and the pasts of a and b . In the causal set the Alexandrov sets are replaced by intervals and volumes, $V[\cdot]$, are replaced the number of points, $N[\cdot]$.

distance between a and b in d dimensions (equivalently, if the spatial distance is much smaller than τ_c), the Minkowski distance between a and b in $d + 1$ dimensions is given by

$$d_{\mathbb{M}^{d+1}}(a, b) = \frac{2}{c_d} \lim_{\tau_c \rightarrow \infty} \tau_c (1 - \mathcal{O}_{\mathbb{M}^{d+1}}(a, b)). \quad (2.46)$$

Here c_d is a dimension dependent constant given by

$$c_d = \frac{d+1}{\sqrt{\pi}} \frac{\Gamma(\frac{d}{2})}{\Gamma(\frac{d+1}{2})}.$$

From the figure it can be seen that as c is taken lower (τ_c is made bigger) $V[C]$ would be much bigger than $V[A]$ (think of a and b being close together and c being far below, then $I(a, c)$ and $I(b, c)$ would almost overlap, i.e., $I(a, c) \simeq I(b, c)$, thus $V[A] = V[I(a, c) \setminus I(b, c)] \ll V[I(a, c) \cap I(b, c)] = V[C]$). Thus, the causal overlap as defined in Eq. 2.45, $\mathcal{O}_c(a, b) \rightarrow 1$ as $\tau_c \rightarrow \infty$, and the distance in Eq. 2.46 remains bounded.

As an example, if two test particles are initially separated by ϵ in the x -direction in a plus polarized plane gravitational wave (GW_+), then we can set up coordinates in which ignoring first and higher order corrections in h_0 (u_α is being multiplied with the Riemann tensor which is already first order in h_0), $\xi^\alpha = (0, \epsilon, 0, 0)$ and the tangent to the geodesics can be taken to be $u^\alpha = (1, 0, 0, 0)$, so to first order in h_0 the deviation equation in the TT gauge reduces to [22]

$$\begin{aligned} \frac{d^2 \xi^x}{d\tau^2} &= \frac{d^2 \xi^x}{dt^2} = \epsilon R^x{}_{00x} = -\epsilon C^+_{x0x0} \\ &= -\frac{1}{2} \epsilon \omega^2 h_0 \cos \omega(t - z). \end{aligned} \quad (2.47)$$

where C^+_{x0x0} was introduced in Eq. 2.27. So, by finding the separation estimates we should be able to numerically find a cosine form for the acceleration of the separation.

There are a few problems that we anticipate in implementing this proposal. The first one is to find the correct pairs of points along the chains (geodesics) that are “simultaneous” at each time step between which to find the spatial distance. This can either be done by first identifying successive inextendible antichains, the analogs of the successive spatial hypersurfaces, one at each time step, $\Sigma_{t_1}, \Sigma_{t_1+\Delta t}, \Sigma_{t_1+2\Delta t} \dots \Sigma_{t_2}$, and then picking points that are found at the crossing of these antichains and the two chains representing the neighboring geodesics. Or one could ignore the antichains and just count going up a link on each chain as an equal time step. The problem with the former is that it is algorithmically hard to identify successive inextendible antichains, and with the latter is that if the two chains representing the geodesics have an unequal number of links then each link cannot

be thought of as an equal time step since the chains are supposed to have an equal proper length.

The second problem we anticipate is extending the results Boguñá et al [4] which are strictly true for causal sets embedded in Minkowski spacetime, but they claim that their results can be extended to curved spacetimes as well. Finally, we might encounter numerical limitations, specifically that the method proposed in [4] gives good estimates of spatial distance for points that have a large shared past in the causal set, thus the cardinality of the causal sets we work with has to be quite large, but such causal sets are time intensive to generate due to the n^2 dependence of the comparisons between points to determine all the relations between sprinkled points.

2.5 A possible extension: pp-waves

We would like to repeat the process of sprinkling for a class of spacetimes called pp-wave spacetimes. They too represent waves in spacetime, but instead of the wave being a perturbation on top of Minkowski spacetime they have the important feature of being *exact* solutions of the Einstein equations in vacuum. They have the following defining properties ([25], p. 244):

$$R_{\alpha\beta} = 0, \quad k_{\alpha;\nu} = 0, \quad k_\alpha k^\alpha = 0. \quad (2.48)$$

Here, k_a is a null wave-vector, which in the case of the linear waves of the previous sections was $(\omega, 0, 0, \omega)$. The second requirement means that these waves are *plane-fronted waves with parallel rays (pp-waves)*. From [21], “A plane wave has a planar wavefront (roughly, the spacelike codimension-2 hypersurface orthogonal to the wave vector), but in general these parallel waves can have curved wavefronts. Although to obtain wavelike behaviour it is not necessary to demand the wavefront be flat, it is standard in the field to make this restriction.” Metrics that satisfy these properties can be written (locally) in a simplified

form as

$$ds^2 = dx^2 + dy^2 + 2 du dv + H(x, y, u) du^2, \quad (2.49)$$

where H must satisfy the condition

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) H(x, y, u) = 0$$

in order to satisfy Einstein's equations.

We anticipate some problems, however. H necessarily contains singularities [25] because no solution of the condition given above is regular over the whole (x, y) plane. So, we will have to see how to deal with these. Moreover, it is not clear whether we will be able to find relations between points using the current method because although it is true that all spacetimes have the property of having convex neighbourhoods around each point we cannot say if the ones in these spacetimes will be large enough to show the effects of curvature and still remain convex. Moreover, because we won't have a parameter like h_0 to expand in, we don't know if we will be able to exactly solve the geodesic equations like we were able to for the linear waves. So, we might have to *come up with a new way of finding relations*. Lastly, we would have to see if the proposal we make for finding the discrete Weyl tensor for linearized plane waves would also work for these spacetimes.

Chapter 3

DYNAMICS

The second project on which I would like to work pertains to the other field of research in CST called “dynamics”. Dynamics means the principles or laws that govern the structure or evolution of the causal set [24]. Ultimately, the dynamics for causal sets is expected and hoped to be quantum mechanical in nature, most likely in the form of “sum-over-histories”, where a single causal set is considered one history. However, realizing it is difficult on at least two fronts. First, the definitions of the initial and final states that the histories in the sum must have are not agreed upon because these states are supposed to be the analogs of spatial hypersurfaces and those are difficult to define in causal sets such that they contain as much information as their continuum counterparts. Second, even if these states are adequately defined, the current proposals for causal set actions are not general enough. In light of such difficulties, the causal set community thought it prudent to develop classical dynamics for causal set. Such dynamics are not expected to be the final ones since they don’t have a quantum character; however if some of them are able to reproduce the predictions of general relativity in the continuum limit then the final dynamics of CST must at least have these models in its continuum limit.

This in itself is enough reason to pursue a classical dynamics; however, it so happens that the primary proposal for classical dynamics of CST, called classical sequential growth (CSG), which views the evolution of a causal set as a growth process where a new element is added to the causal set such that it is not to the past of any of the existing elements, is successful on other fronts and thus contains lessons for quantum dynamics. Briefly, these are the facts that it is unexpectedly able to overcome the problem of the overabundance of non-manifoldlike causal sets in the sample space, Ω , of past-finite causal sets where the

dynamics (quantum or classical) of CST are expected to play out.¹ Moreover, it satisfies two physically motivated principles: i) *discrete general covariance* and ii) *Bell causality*. Roughly, the former means that the transition probabilities cannot depend on the labelling assigned to the elements of a causal set, which is analogous to the requirement of coordinate invariance in GR, and the latter “captures the intuition that a birth taking place in one region of the causet cannot be influenced by other births that occur in regions spacelike to the first region.” [23]

Despite these successes, the CSG models have a serious shortcoming, which is that at least some of the models within the CSG framework do not produce manifold-like spacetimes easily/often, nor are they able to maintain through the growth process the manifold-likeness of causal sets that are large and manifold-like to begin with.² This means that CSG models on their own are not sufficient as classical growth models. What one would like, e.g., is for a causal set that is initially embeddable in Minkowski spacetime to grow in a way that it continues to be embeddable, albeit in a larger patch, of Minkowski spacetime.

3.1 Our idea

The goal of our project, thus, is to modify the CSG framework so that the evolved causal sets continue to be embeddable. We propose to do this by making *the transition probabilities of CSG depend in some way on the value of the discrete action for causal sets*.

The action is a natural choice for use in formulating dynamics because unlike the Lagrangian and Hamiltonian it is defined over the bulk of spacetime in both traditional for-

¹This means that if one were to construct causal sets using n elements most of them will have just three layers in which all the elements are arranged, called Kleitman-Rothschild causal sets. CSG models, which are all characterized by some specifiable transition probabilities $\{t_n\}$ from a causal set of cardinality n to another one of $n + 1$, are able to suppress the tendency for a causal set to randomly grow into a KR causal set.

²This has been shown recently by He Liu, a graduate of our group.

malims and CST. This is not true for the Lagrangian or the Hamiltonian, or geometric quantities like curvature and the metric in general, as they cannot be defined at a *single* point or hypersurface (maximal antichain) of a causal set like they can be in continuum theories. This is because the *only* information in a causal set is in the *relations between elements* (without the relations the causal set is a collection of disjoint elements), and since a point or an anti-chain don't contain any relations by definition, they don't hold any information in and of themselves. Thus, at best one can associate the *average* of a quantity in a neighbourhood of a point (or an antichain) with the point (or the antichain) itself by summing over the neighborhood. This makes the definition of these quantities inherently non-local and one cannot use traditional formalisms for dynamics that employ them *without some adaptation* (this motivates our exploration of non-local theories of Sec. 3.4.3). Thus, our hope is that due to the similarity of the definition of actions in traditional formalisms and causal sets, it will be *easier and more direct* to take inspiration from and adapt these formalisms for formulating dynamics for causal sets.

There are a couple of proposals for discrete actions for causal sets, which though restricted in various ways nonetheless approach the value of the Einstein-Hilbert action in the continuum limit. An example is the “chain” action which uses the expression for the Ricci scalar Eq. 2.43 and gives the Einstein-Hilbert action $\int R(x) \sqrt{-g(x)} d^4x$ as $\sim R(x_0) \cdot V_0 = R(x_0)n/\rho$ for an Alexandrov set of volume V_0 small enough for the curvature to be constant (here, x_0 is the minimal element of the Alexandrov set):³

$$S_{c_3} = 36 \left(\frac{4c_3}{N(n-1)} - 1 \right). \quad (3.1)$$

With this background in place, we can discuss what progress we have made. We don't yet have a concrete proposal for how to link the transition probabilities to the action. This

³This expression is not defined for a single causal set. The c_3 is actually $\langle c_3 \rangle$, i.e., the mean number of 3-chains calculated for an ensemble of sprinklings into the same spacetime. Whether the average for the ensemble can be replaced by the value of c_3 for a particular causal set is an open question.

is because of the conceptual difficulties of the problem, some of which are explored in the next section. Our strategy has been to focus on continuum-inspired dynamics that involve the action in some significant way. The most important case is that of *Weiss variation*, where one allows for the variation of the configuration variables *at the boundaries*, along with the familiar variation of the path in the bulk. Our hope was to find a condition on the variation of the action, δS , in terms of the variations of the configuration variables at the boundaries. Currently we are struggling with associating appropriate causal set structures to the continuum variables. The other major area of our focus is *theories with non-local Lagrangians*, i.e., the Lagrangian depends on the values of the generalized coordinates at future and past times. We are pursuing these because, as has been described above, the Lagrangian or Hamiltonian for a causal set (if there is hope of finding them) will have to be non-local. Our goal is to see if anything relevant for the dynamics of causal sets can be learnt from the dynamics of these theories. For both of these areas, and others that we have looked at, our work at this stage remains exploratory, and because of its exploratory nature this project is not our primary focus. My plan is to work on it after finishing the project on the Weyl tensor.

The structure of the rest of the chapter is as follows. I will state a few facts about variations in causal sets that shed more light on why dynamics for causal sets are hard to develop. Then, I will return to the case of viewing classical dynamics for causal sets as growth by addition of elements and describe what it means to add an element to a causal set and how we would like the dynamical law to tell us which of the existing elements the new element should be related to. And finally I will describe some key findings from our study of different ideas from continuum theories. Some of these findings are not directly relevant to our project but are included as a way of showing how in each approach there are unexpected and interesting ideas.

3.2 Difficulties in formulating classical dynamics

As has been said, a major source of difficulty in coming up with an evolution formalism for causal sets is that a Lagrangian or a Hamiltonian will have to be non-local, and in any case we do not have any proposals for them yet. However, something to keep in mind is that even if one were to propose a non-local Lagrangian, it would be extremely challenging to use calculus of variations to find the equations of motion. This is because it is difficult to vary a causal set for reasons given below.

1. At the smallest scale the least change one can make to a causal set is changing a binary relation between two elements. Recall that a causal set can be fully described by the links matrix, L . The smallest change then is creating or destroying a link, i.e. $0 \rightarrow 1$ or vice versa in the links matrix. A binary change is obviously not an infinitesimal change. Since the typical condition on the action is $\delta S = 0$ only for an *infinitesimal* change in the variables it is not clear what the condition on the action should be in the case of a discrete change.
2. The other, and more significant, reason is that most of the times we are forced to make multiple non-local changes to a causal set even if we try to change a single relation. This is due to the preservation of *transitivity* between the points in consideration and other points. For example, if $1 \prec 2 \prec 3$, and we remove the relation between points 1 and 3 ($1 \not\prec 3$) then the remaining relations, $1 \prec 2$ and $2 \prec 3$, obviously contradict this move. Thus, we are forced to make more changes. This not only makes *single* changes hard to do, it also makes solely local changes hard to implement. Physically this issue is illustrated by the following reasoning. If we imagine removing a link between two points, in the continuum limit that means separating two nearby points in a manifold. Which means also to separate two regions. To prevent that we will either have to change other relations which might even be far away or declare that there is a hole in the manifold.

In both cases the change is not small.

3.3 Growing a causal set by one element

As was said in the Introduction, a causal set can be fully described by what is called a links matrix, $\mathcal{L}$. If we arbitrarily label all the elements of a causal set 1 to n then the links matrix will be an $n \times n$ matrix with the ij -th element being 1 if the element $i \prec^* j$ and zero otherwise.⁴ The changes that take place in a causal set, $\mathcal{C} \rightarrow \tilde{\mathcal{C}}$, due to the addition of one element are that (i) the cardinality of $\tilde{\mathcal{C}}$ is one larger than that of $\mathcal{C}$, consequently the links matrix, $\mathcal{L}_{n \times n} \rightarrow \tilde{\mathcal{L}}_{(n+1) \times (n+1)}$, grows by a row and a column, and (ii) a choice of a precursor set c_i is made for the new element, which is the set of elements the new element is to the future of; this is the same as saying the entries of the new row and column of the links matrix are decided. A new element can be in the future of different subsets of existing causal set, $c_i \subset \mathcal{C}$, called *precursor* sets. Crucially, choice of each precursor set will produce a different causal set $\tilde{\mathcal{C}}$, so each birth of a new element opens the possibility for the existing causal set to *transition* into multiple new causal sets. Consequently, one can imagine a space of causal sets with subsets that all have the same cardinality but the relations among their elements are different and a sequence of births is a sequence of transitions in this space from one causal set to another. Part of the sample space of causal sets grown from a single element is given in Fig. 3.1

Each choice of the precursor set affects the continuum approximation (faithful embedding described previously) of the grown causal set. We want the choice to be such that the grown causal set is faithfully embeddable into the evolved version (larger region) of the classical manifold. For static spacetimes this would mean that the grown causal set maintains the curvature of the spacetime, e.g., a causal set initially faithfully embeddable

⁴Recall that $\prec^*$ is called a link, where $i \prec^* j \implies i \prec j$ and $I(i, j) := \{r \mid i \prec r \prec j\} = \emptyset$, which denotes that i and j have no element between them.

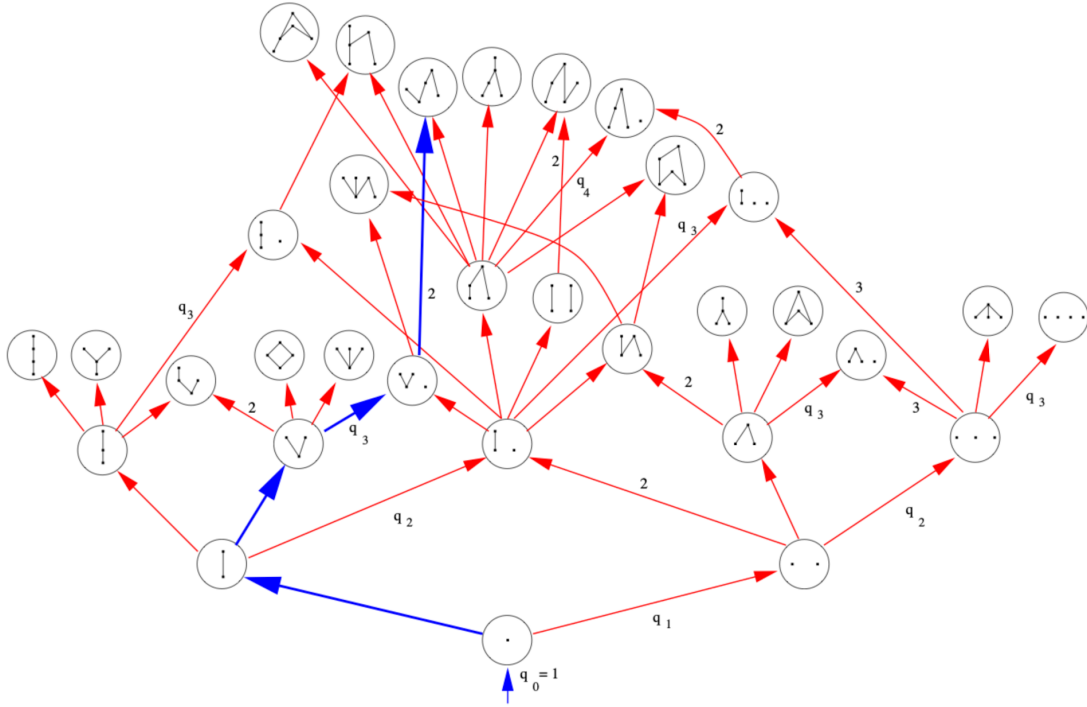


Figure 3.1: Limited sample space for causal sets grown from a single element and a path in it. Technically a Hasse diagram of Hasse diagrams. Only 8 out of 63 five element causal sets are shown. The goal of a dynamical law is to dictate the path taken through the sample space. [Source: the image is taken from [20]]

in a patch of Minkowski spacetime should continue to be so for a larger patch during the growth. Whereas for dynamic spacetimes the growth process should ensure faithful embeddability in the dynamically evolved patch of the spacetime, e.g., for the GW_+ of the previous chapter if our causal set has been sprinkled in a region that ends at a crest of the wave in the spacetime, after a short growth interval the grown causal set should be embeddable in a slightly larger region that ends at a phase of decreased amplitude. So, the correct dynamics is supposed to tell us which precursor set c_i to choose.

An important note is due here about the fact that the choice for the c_j does not have to be unique for the evolved causal set to continue to be embeddable. This is because multiple causal sets can be faithfully embedded in the *same* spacetime. The easiest way to see this is to recall the sprinkling process introduced in chapter two — it is a way to generate

causal sets from a given spacetime, which in our case was the GW_+ , using *probability distributions*. Therefore, running the sprinkling algorithm generates a different causal set each time. This principle would apply to any spacetime being sprinkled into, hence there will be multiple causal sets that can be embedded into the evolved spacetime under consideration. Thus, *the dynamical law does not have to produce a unique future causal set*.

For reference, CSG models have the following form for the transition probabilities, i.e., the probability to go from an $\mathcal{C}_n$ to $\tilde{\mathcal{C}}_{n+1}$ [20]:

$$\alpha_n = \frac{\sum_{l=m}^{\omega} \binom{\omega-m}{\omega-l} t_l}{\sum_{j=0}^n \binom{n}{j} t_j} \quad (3.2)$$

where m is the number of maximal elements in the precursor set (an element of $\mathcal{C}$ is maximal iff it is to the past of no other element) and ω is the size of the entire precursor set, and the $\{t_n\}$ are freely specifiable constants. An example choice for the $\{t_n\}$ is $t_n = t^n$, which is the case of transitive percolation, where every new element independently forms a link with each existing element with the probability p that can be specified, then $t = p/(1-p)$.

3.4 Things we have explored and our reasons

We have been studying formalisms that share features with CST, however, one of our hopes is to develop results that are relevant to the classical dynamics of continuum theories in general, so in our discussion below not all aspects are directly relevant to causal sets. We will discuss: Weiss variation, Hamilton-Jacobi theory, the derivation of the integral version of Noether's theorem, and dynamics for non-local Lagrangians. For each of them, we will explain why we became interested in them and what the current challenges are.

3.4.1 Weiss-variation

The idea of Weiss variation is to allow for the variation of the configuration variables *at the boundaries*, along with the familiar variation of the path in the bulk with fixed end

points. The key result of this formalism is that to get the Euler-Lagrange equations we have to change the variation condition. Taking the example of a system described by one generalized coordinate $q(t)$: instead of requiring $\delta S = 0$ on the physical path between times t_2 and t_1 with fixed $q(t_2)$ and $q(t_1)$, we need the condition

$$\delta S = (p\Delta q - H\delta t) \Big|_{t_1}^{t_2}, \quad (3.3)$$

where $p := \partial L / \partial \dot{q}_a$ is the conjugate momentum, L is the Lagrangian, $H(t) := p(t)\dot{q}(t) - L(t)$ is the Hamiltonian, $\Delta q(t) := \dot{q}(t)\delta t_i + \delta q(t)$ is the *total* variation in the end-point values of q which can be different at t_2 and t_1 , and $\delta q(t)$ and $\delta t(t)$ are the small arbitrary variations of the end-points under ones control [8]. That is, the physical path is that around which the variation of the action and boundaries depends up-to first order *only* on how much the end-points of the varied paths are shifted.

To see why we became interested in this recall how the problem of how to grow a causal set by the addition of elements is the problem of selecting precursor sets c_i for each new element added. Our idea was that each choice of the precursor set will effect the value of the action differently and since we know how to find the action for a limited class of causal sets we could try to find a condition on the change in the action δS that would allow us chose between the various subsets for the new element to be related to. And our hope was that Eq. 3.3 adapted appropriately, either for n generalized coordinates or for a classical field (including GR [8]), would provide this condition (see the footnotes below).

Something that encouraged us to pursue our line of reasoning is that there is an implicit similarity between Weiss variation and our idea which is that both assume that the variation is taking place around the physical path/causal set. Unlike Eq. 3.3, a variation around a non-physical path produces a longer expression for δS :⁵

$$\delta S = \int_{t_1}^{t_2} dt \left(\frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} \right) \delta q + [p(t)\Delta q(t) - H(t)\delta t_i] \Big|_{t_1}^{t_2} \quad (3.6)$$

⁵The analog expression for a collection of *classical fields*, φ^A , in a region $\mathbf{W} \in \mathcal{M}$ of a Lorentzian

The integrand of the first term is the one found in the Euler-Lagrange equation and is zero for the physical path between t_1 and t_2 . So only for a physical path does Eq. 3.6 reduce to Eq. 3.3. The CST equivalent of a physical path would be a causal set which is faithfully embeddable in a corresponding spacetime, then adding elements to it would be akin to varying the boundaries of a physical path at t_2 .

Before applying our idea to causal sets we first tried to apply it to classical mechanics itself to see if Eq. 3.3 can be used as condition for figuring out what $q(t)$ will be beyond t_2 given $q(t_2)$ is already on the physical path. A visual representation of this question is drawn in Fig. 3.2:

Unfortunately, this doesn't seem possible because Eq. 3.3 does not provide a fixed value for δS like the condition $\delta S = 0$ does. This means $\Delta q_a(t)$ and $\delta t(t)$ can be anything we choose in Eq. 3.3,

$$\delta S = (p\Delta q - H\delta t)|_{t_1}^{t_2}, \quad (3.3)$$

and the value of δS will change depending on our choice. All the equation demands is that manifold with boundary $\partial\mathbf{W}$ is given by: [8]

$$\delta S := \int_{\mathbf{W}} \left(\frac{\partial \mathcal{L}}{\partial \varphi^A} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi^A)} \right) \right) \delta \varphi^A d^4x + \int_{\partial \mathbf{W}} (P_A^\mu \Delta \varphi^A - \mathcal{H}^\mu{}_\nu \delta x^\nu) d^3 \bar{\Sigma}_\mu, \quad (3.4)$$

where $\mathcal{L}(\varphi^A, \partial_\mu \varphi^A, x^\mu)$ is the *Lagrangian density* defined at each spacetime point, $P_A^\mu := \partial \mathcal{L} / \partial (\partial_\mu \varphi^A)$ is called the *polymomentum*, $\mathcal{H}^\mu{}_\nu := P_A^\mu \partial_\nu \varphi^A - \delta_\nu^\mu \mathcal{L}$ is called the *Hamiltonian tensor*, and $d^3 \bar{\Sigma}_\mu$ is the directed surface element on $\partial \mathbf{W}$.

The $\mathcal{H}^0{}_0$ component is defined as the *Hamiltonian density*, $\mathcal{H} := \mathcal{H}^0{}_0 = \pi_A \dot{\varphi}^A - \mathcal{L}$, where the overdot denotes a derivative w.r.t. to the time coordinate x^0 , thus $\dot{\varphi}^A := \partial_0 \varphi^A$ and $\pi_A := P_A^0 = \partial \mathcal{L} / \partial \dot{\varphi}^A$ is called the *conjugate field momentum*. The field Lagrangian and Hamiltonian can be obtained from the respective densities:

$$L[\varphi^A, \dot{\varphi}^A; \Sigma_t] = \int_{\Sigma_t} \mathcal{L}(\varphi^A, \partial_\mu \varphi^A, x^\mu) d^3y, \quad H[\varphi^A, \pi_A; \Sigma_t] := \int_{\Sigma_t} (\pi_A \dot{\varphi}^A - \mathcal{L}) d^3y. \quad (3.5)$$

Here Σ_t is a hypersurface of constant t with spatial coordinates y^i on it, and the semicolon in $L[\varphi^A, \dot{\varphi}^A; \Sigma_t]$ (and H) shows that we are considering fields restricted to the hypersurface, $\varphi^A(y^i)|_{\Sigma_t}$ and $\dot{\varphi}^A(y^i)|_{\Sigma_t}$.

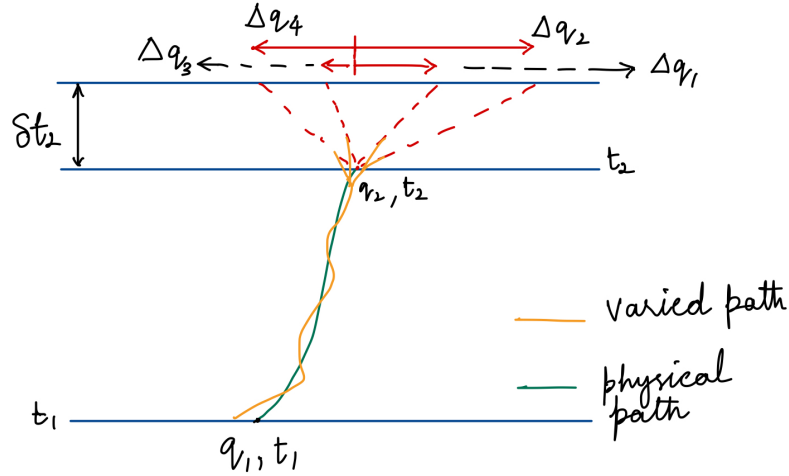


Figure 3.2: Can we come up with a condition on the action or its variation/change such that it tells us which of the Δq_i s is allowed in the time interval δt_2 ? δt_2 has been exaggerated for clarity.

there should be no contributions to δS from the bulk variation, whereas the contributions from the boundary variations should appear in the combination given by the equation. This is precisely what we don't want. We want to know what $q(t)$ will be at $t_2 + \delta t_2$ and beyond.

In the context of causal sets, however, we are hopeful that this approach might still work. The first step would be to specify the relationship between δq and δt and the changes that take place in the causal set due to the growth. If we take the naive approach that $\delta t(t_2)$ is proportional to the cardinality of the subset c_i and assume $\delta q = 0$, we will see that there is degeneracy in the choice of c_i because there can be many c_k with the same cardinality. The important idea is that the value of the action for causal sets grown using different choices for c_i will be different, which we can find using their links matrices. Now, if one were to use the GR version of Eq. 3.3 for some chosen value of $\delta t(t_2)$ to find δS_{theory} one can imagine that only a few c_j (or even a single one) will produce a δS that is close (or equal) to the theoretical value. If we choose only these c_j then we have a way to select between all possible c_i s. However, we cannot say more about this idea since it is a recent development

and most of its details need to be worked out.

3.4.2 Noether's theorem's integral version

It turns out that unlike most Quantum Field Theory textbooks some authors like Goldstein [10] and Bogoliubov and Shirkov [3] define a continuous symmetry for a classical field as the *invariance of the action*, $\delta S = 0$, under an infinitesimal transformation of the field variables and/or the spacetime coordinates, and their expression for the change in the action due to the transformations is exactly the same as that for Weiss variation (compare Eq. 2.25 of [8] with Eqs. 13.142 and 2.5 of [10] and [3] respectively). That is, taking the example of Eq. 3.3,

$$\delta S = (p\Delta q - H\delta t) \Big|_{t_1}^{t_2}, \quad (3.3)$$

both would agree on the expression for δS (cf. Eq. 13.158 of [10]) but if δq or δt are to be symmetry of the system then the R.H.S. would be zero according to the former. Moreover, equating the R.H.S. to zero with $\Delta q = 0$ and $\delta t_2 = \delta t_1$ immediately implies that H is constant. This is an example of the famous Noether's theorem.

The relevance of this form of the Noether's theorem for us is that it provides a constraint on the change in the action of a causal set due to certain kinds of growth without even comparing with GR as in the previous section if we suppose that the relationship between the conserved physical quantities and their respective symmetries holds for causal sets as well. Once again, however, the idea has not been implemented yet and the details need to be worked out.

A few things that need clarification are as follows. First, we need to decide feature of the causal set growth do we associate with the symmetries δt or δq . Similarly, we need to propose the analogs of p and H in a causal set. More needs to be understood about the details of Noether's theorem. For example, we need to confirm whether Noether's theorem can be applied to the case where there is no variation at t_1 and the spatial boundaries, and

the only variation is the t_2 surface since we are postulating dynamics to mean growth at the future boundary of a causal set. Moreover, there are conflicting opinions about the necessary conditions for the Noether's theorem to hold, e.g. some resources say that a Noether symmetry is a symmetry on all paths, not just the classical path. Addressing these issues would be the first step towards using conserved quantities for causal set dynamics.

3.4.3 Dynamics for non-local Lagrangians

The idea is to study continuum theories whose Lagrangians depend on the values of the generalized coordinates at future and past times. The motivation for doing so comes from the fact that causal sets have non-local features, as mentioned earlier. As explained there, geometrical quantities cannot be defined at a point in a CS. Thus, to find the Einstein-Hilbert action for a causal set, we not only have to integrate over a spacetime region once, as is typically done, we need a *second* integral over the whole or part of the region to first estimate the value of $R(x^\mu)$ at each point. Moreover, this neighborhood can contain points that are spatially very far away (even infinitely) from the point due to local finiteness of density of points, which means that the domain of the second integral should technically be the whole spacetime. This is a second way in which non-locality appears in CST. It happens in causal sets since it is possible for a very far away point to form a spacetime interval with the point in consideration such that there happen to be no other points in it. This does not happen in the continuum because there are always more points between any points by the definition of continuity.

We have so far been interested in three non-local ideas. First, theories that have Lagrangians of the type given in Llosa and Vives [13]. The simplest example they cover is the following Lagrangian:

$$L(t) = \frac{1}{2}\dot{q}^2(t) - \frac{1}{2}q^2(t) + \frac{g}{4}q(t) \int_{\mathbb{R}} dt' q(t') e^{-|t-t'|} \quad (3.7)$$

in which the last term makes $L(t)$ depend on the values of $q(t')$ at past and future times t' , albeit with an exponentially decaying contribution. The Euler-Lagrange equations are schematically the same:

$$\int_{\mathbb{R}} dt \frac{\delta L(t)}{\delta q^l(t')} = 0, \quad (3.8)$$

one for each generalized coordinate $q^l(t)$. $\frac{\delta}{\delta q^l(t')}$ is the functional derivative w.r.t. $q^l(t')$. This way of writing the Euler-Lagrange equations works for the usual local Lagrangians as well. For this particular Lagrangian the E-L eq. become

$$-\ddot{q}(t) - q(t) + \frac{g}{2} \int_{\mathbb{R}} dt_1 q(t_1) e^{-|t-t_1|} = 0. \quad (3.9)$$

Second, we want to investigate the continuum versions of existing causal set actions. One such action is derived from results of the example quoted in Sect. 2.3.2. In 2D, for a region with constant R in an Alexandrov interval A_0 , one can use the expression for $\langle c_3 \rangle$ to define

$$S_3[g] = \frac{144}{V_0^2} \int_{A_0[g]} \left(\int_{A_1[g]} \sqrt{-g} d^2 x_2 \right) \sqrt{-g} d^2 x_1 - 36 \quad (3.10)$$

Here we see the two integrals mentioned before. $S[g]$ means that the action depends on the metric. I have not been able to investigate this action further. One problem I am facing is how to define the dynamical variables and how to gauge the seriousness of the fact that R is assumed constant in it. This is on the agenda for future work.

Third, we want to explore Synge's world function which is the half the geodesic distance between two points that lie on a geodesic γ [19]:

$$\sigma(x, x') = \frac{1}{2}(\lambda_1 - \lambda_0) \int_{\lambda_0}^{\lambda_1} g_{\mu\nu}(z) t^\mu t^\nu d\lambda, \quad (3.11)$$

where the geodesic has components $z^\mu(\lambda)$, λ is the affine parameter along the geodesic and $t^\mu = dz^\mu/d\lambda$ is the tangent vector to the geodesic. Its significance lies in the fact that it is a nonlocal function of two points in the manifold and bears resemblance to the continuum

versions of the actions in CST, in whom the integrand is a function over a *region* in the manifold.

3.4.4 *Training a machine learning algorithm on large causal sets*

Finally, it is worth exploring the possibility of growing a causal set using a machine learning (ML) algorithm. The idea is to sprinkle thousands of large sized causal sets (cardinality $\sim 10^4$) in a simple spacetime like Minkowski and training a machine learning algorithm on them in order for it to recognize new causal sets that are also embeddable in Minkowski and, crucially, which don't have the same cardinality as the training set. Our hope is that if the latter can be done then there is a possibility to develop a further algorithm that uses the aforementioned ML algorithm and takes as input a Minkowski embeddable causal set of certain cardinality and returns a causal set that is the same as the input but with new elements added to it such that it is also Minkowski embeddable. This would be the same as having grown the causal set. The subsequent challenge would be to translate into physical terms why the algorithm succeeds in correctly growing the causal set.

This is the least explored idea out of all of the previous ones however, it seems one of the most promising ones given that causal sets are naturally represented as sparse square matrices and the fact that ML models employ matrix operations extensively.

3.4.5 *Hamilton-Jacobi theory*

The Hamilton-Jacobi equation is a single partial differential equation for the variable $S(q, \alpha, t)$ that is interpreted as the classical action, in terms of the Hamiltonian H :

$$H\left(q, \frac{\partial S}{\partial q}, t\right) = -\frac{\partial S}{\partial t}. \quad (3.12)$$

There are two other relations that are implicit which, together with the previous equation, form the “Hamilton-Jacobi *formalism*”:

$$p = \frac{\partial S}{\partial q} \quad (3.13)$$

and

$$\beta = Q = \frac{\partial S}{\partial \alpha} \equiv \frac{\partial S}{\partial P}. \quad (3.14)$$

Here, (P, Q) are canonically transformed variables in which the original Hamiltonian H is transformed to a new one K such that $K = 0$, which implies from Hamilton’s equations that P and Q must be constants; α and β are constants representing values of (P, Q) that can be found in terms of the initial conditions; and p is the conjugate momentum of q .

The strength of the Hamilton-Jacobi formalism is not the reduction labor. Its strength is that it can reduce the whole problem, regardless of the number of generalized coordinates, of solving $2n$ first-order differential equations to a *single* first-order PDE. It, moreover, *allows the dynamical equations to be formed in terms of the derivatives of the action*. Our hope was to adapt $\partial S/\partial t$ and $\partial S/\partial q$ for the known actions for causal sets and work with the special case in which the Hamiltonian was zero for the causal set. However, we stopped after a while because we could not find counterparts of t and q for the causal set actions, nor fully understand what would be the implications of a causal set with a vanishing Hamiltonian.

An interesting discovery while studying this topic was that in regular classical theory the action can have different functional forms for the same system and the *same variables*. For example, the action for a simple harmonic oscillator can be written for the variable $q(t)$ representing displacement as both:

$$S = \frac{1}{2}q\sqrt{2\alpha - \omega^2 q^2} + \frac{1}{2\omega}\sqrt{2\alpha} \tan^{-1}\left(\frac{\omega q}{\sqrt{2\alpha - \omega^2 q^2}}\right) - \alpha t \quad (3.15)$$

and

$$S' = \frac{\omega}{2 \sin \omega t} [(q^2 + \alpha'^2) \cos(\omega t) - 2q\alpha']. \quad (3.16)$$

The difference in each is due to the physical meaning of (α, β) . In the former $(\alpha, \beta) \rightarrow$ (total energy, initial phase/ ω) and in the latter $(\beta', \alpha') \rightarrow$ (momentum, position).

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